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INDUSTRIAL CONCENTRATION, PROPERTY VALUES, AND MUNICIPAL BOND
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Kenneth R. Ahern

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ABSTRACT

This paper shows that the industrial composition of a city's local economy affects its municipal borrowing costs. In a panel of 1,177 U.S. cities from 2005 to 2022, greater sectoral concentration magnifies default risk and raises bond spreads, especially for cities dominated by industries associated with low property values. Instrumental variables exploiting national sector-employment trends and regional house-price variation support a causal interpretation. A calibrated model of city default suggests that the observed spread effect understates the gross risk created by concentration, because higher concentration can generate agglomeration benefits that reduce spreads, especially for high-property-value cities.

Kenneth R. Ahern
University of Southern California
Marshall School of Business
and NBER
kenneth.ahern@marshall.usc.edu

I. INTRODUCTION

Although public attention often centers on the policies of the federal government, many of the most tangible public services are delivered at the city level, including roads, sidewalks, police, and fire protection. Because cities rely on debt to finance these services, their borrowing costs have meaningful economic consequences. Higher spreads increase infrastructure costs, constrain capital investment, and lower the quality of services for residents (Adelino, Cunha, and Ferreira, 2017). Yet recent data indicate that the fiscal health of cities is deteriorating: since 2021, expenditure growth has outpaced revenue growth for the average U.S. city (National League of Cities, 2025). This imbalance is reflected in the growing number of cities declaring fiscal emergencies, from Los Angeles to Chester, Pennsylvania. These trends raise a fundamental question: what determines city borrowing costs?

In this paper, I argue that an important determinant of a city’s borrowing costs is the industrial composition of its local economy. In a distance-to-default framework (Merton, 1974; Leland, 1994), industrial composition affects spreads through two dimensions: the concentration of the city’s industrial base and the level of its latent fiscal capacity, defined as the untapped potential to raise additional revenue. Cities anchored by high-wage industries tend to have higher property values and therefore greater latent fiscal capacity, which reduces borrowing costs.

The effect of industrial concentration, however, is ambiguous because it operates through two opposing forces. Through a *risk channel*, concentration raises revenue volatility: cities supported by a more concentrated set of industries have greater exposure to sector-specific shocks. Through an *agglomeration channel*, concentration strengthens the local tax base by generating productivity gains through knowledge spillovers, labor-market pooling, and input sharing (Marshall, 1890; Duranton and Puga, 2004; Rosenthal and Strange, 2004). Which force dominates is an empirical question. The framework also predicts that fiscal capacity and concentration interact: concentration should affect borrowing costs more strongly in cities with lower latent fiscal capacity, because revenue volatility matters most when cities are closer to distress.

To test these hypotheses, I construct a city-year panel of 1,177 cities from 2005 to 2022 by matching secondary-market municipal bond trading data from the MSRB to employment data from the U.S. Census Bureau’s LODES program. The employment data provide sector-level job counts at the city level, from which I compute the Herfindahl–Hirschman Index (HHI) of employment concentration across 20 NAICS supersectors. The bond data provide

par-value-weighted average yields, from which I compute yield spreads against a contemporaneous, maturity-matched benchmark curve constructed from the universe of AAA-rated municipalities.

In the first set of tests, I isolate the effect of HHI and property values on credit spreads by controlling for city and year fixed effects plus a host of time-varying explanatory variables. The results show that employment concentration is priced only for cities in the lowest tercile of median property values. Among these cities, a one-standard-deviation increase in HHI is associated with a 4.2 basis point increase in spreads, relative to a sample mean of 21 bps. This is a sizable economic magnitude, equivalent to roughly seventy percent of the effect of a one-percentage-point increase in the unemployment rate. Among cities in the highest tercile of property values, the effect is economically and statistically indistinguishable from zero. The data therefore identify a heterogeneity in pricing: the risk channel of concentration outweighs the agglomeration channel only where fiscal capacity is limited.

Because HHI and property values are jointly determined by local economic conditions, neither can be treated as exogenous. To provide causal evidence, I develop an instrumental variables design that jointly instruments employment concentration, property values, and their interaction. HHI is instrumented by a Bartik shift-share predictor that interacts predetermined city-industry employment shares with exogenous national sector trends, following Goldsmith-Pinkham, Sorkin, and Swift (2020) and Borusyak, Hull, and Jaravel (2022). Log property value is instrumented by a cumulative housing shock that interacts city-specific housing supply elasticities with regional house-price movements (Guren, McKay, Nakamura, and Steinsson, 2021). The interaction between concentration and property value is instrumented from squares and cross-products of the two base IVs.

The two-stage least squares results provide causal evidence consistent with the OLS findings: increases in HHI cause higher credit spreads, consistent with the risk channel; higher property values cause lower spreads, consistent with the latent capacity channel; and the interaction between concentration and property values is negative, indicating that the concentration effect on spreads is larger for cities with lower property values. This result confirms that the interaction reflects a structural economic relationship rather than selection on observable city characteristics.

Next, I show that the results survive a battery of alternative explanations. First, concentrated cities could have greater sensitivity of local employment to common business-cycle shocks. Controlling for a city-level beta that weights national sector betas by local employment shares (Tuzel and Zhang, 2017), the HHI coefficient and its interaction with property values are qualitatively unchanged across all specifications, ruling out systematic risk as the

driver of the result. Second, municipal bonds are disproportionately held by local retail investors (Schultz, 2012; Babina, Jotikasthira, Lundblad, and Ramadorai, 2021), and concentrated cities could exhibit stronger local bias. I show that the results are robust to both illiquidity and overoptimism caused by local bias. Third, concentrated cities might simply have weaker governance. City fixed effects and the Bartik instrument address this concern. Fourth, state-level institutional constraints such as balanced-budget requirements and bankruptcy authorization could drive spreads, but these effects are largely time-invariant and absorbed by city fixed effects; the results are further robust to state $\times$ year fixed effects.

In the last section of the paper, I estimate a stylized model of city default that decomposes the net effect of concentration identified in the empirical results into separate channels of agglomeration and risk that cannot be directly observed. A city’s industrial composition is modeled as evolving according to a Gibrat process with mean reversion, which jointly determines its employment concentration and property values. Cities collect revenue from property taxes and employment-sensitive sources, face annual shocks that propagate from sector-level growth to city-level revenue through employment shares, and default when the realized revenue shortfall exceeds the city’s fiscal capacity. The calibrated results reveal that the gross risk effect of concentration is several times the observed net effect, with the agglomeration cushion offsetting roughly two-thirds of the gross risk in low-property-value cities and more than fully offsetting it in high-property-value cities. Thus, the flat HHI–spread relationship in wealthy cities arises not because concentration is harmless but because their agglomeration cushion is several times larger than that of fiscally vulnerable cities.

This paper’s primary contribution is to identify a new economic channel influencing municipal borrowing costs: the industrial composition of a city’s local economy. Industrial composition is a foundational feature of every city’s tax base: it shapes both the level of local revenues and their sensitivity to sector-specific shocks. While an earlier literature provides correlational evidence that tax capacity and local fiscal conditions affect borrowing costs (Capeci, 1994), a growing literature shows that municipal borrowing costs also respond to a variety of local shocks and conditions—opioid crises (Cornaggia, Hund, Nguyen, and Ye, 2022), hurricanes (Jerch, Kahn, and Lin, 2023), COVID-driven out-migration (Gustafson, Haslag, Weagley, and Ye, 2025), sea-level rise (Goldsmith-Pinkham, Gustafson, Lewis, and Schwert, 2023), demographic aging (Butler and Yi, 2022), cybersecurity breaches (Curti, Ivanov, Macchiavelli, and Zimmermann, 2024), and state bankruptcy regimes (Gao, Lee, and Murphy, 2019). This paper complements that literature by studying not an external

shock to city governments, but the underlying structure of the local tax base that shapes cities' exposure to such shocks.

This paper is complementary to a related literature that studies how information about a city and its market intermediaries—rather than the underlying local economic environment—shape borrowing costs. Cornaggia, Cornaggia, and Israelsen (2018) exploits the 2010 Moody's rating recalibration to show that credit ratings causally affect municipal borrowing costs, and Cornaggia, Cornaggia, and Israelsen (2023) documents conflicts of interest in rating-agency fees for public finance. Gao, Lee, and Murphy (2020) documents that the depth of local newspaper coverage predicts cross-city spread variation. Ivanov and Zimmermann (2024) document pricing frictions in secondary-market trading, and Ivanov, Zimmermann, and Heinrich (2025) show how disclosure requirements affect municipal outcomes. Whereas these papers examine how the information and intermediary environment surrounding a city's bonds is priced, my paper identifies the effect of the underlying local economic environment itself.

II. HYPOTHESIS DEVELOPMENT

In this section of the paper, I describe a simple, non-structural framework to motivate the empirical analysis. In Section VII, I calibrate a quantitative model of city default. As an overview of the framework, I first establish that municipal bond spreads are determined in large part by default risk from exposure to idiosyncratic shocks. Then, I show in a distance-to-default framework that the relationship between industrial concentration and default risk is theoretically ambiguous: while greater concentration increases a city's exposure to idiosyncratic sectoral shocks, it may also have offsetting positive effects on a city's fiscal capacity through agglomeration benefits. Thus, whether concentration has a positive or negative effect on bond spreads is an empirical question. Finally, the framework shows that concentration has a larger effect on default when latent fiscal capacity is small.

II.A. Municipal Bond Spreads Are Priced on Idiosyncratic Default Risk

Prior research decomposes the determinants of municipal bond spreads into three components: a tax effect, a liquidity premium, and a default premium (Longstaff, 2011; Wang, Wu, and Zhang, 2008; Garrett, Ordín, Roberts, and Suárez Serrato, 2023). In particular, Schwert (2017) estimates that default risk accounts for roughly 74 to 84 percent of the tax-adjusted average spread, with liquidity accounting for the rest. This implies that default risk is the primary driver of bond spreads, but also that it is important to control for liquidity and tax effects.

A second well-documented feature of the municipal bond market is that bond prices primarily reflect idiosyncratic rather than systematic risk. This follows from the local tax-driven structure of the municipal bond investor base. While federal tax exemptions apply broadly, state-level exemptions typically apply solely to in-state bonds, giving investors a strong incentive to hold home-state issues rather than a nationally diversified municipal bond portfolio. Consistent with this, households hold the majority of outstanding municipal debt, either directly or through mutual funds and separately managed accounts (Schultz, 2012). The resulting investor base is fragmented and home-biased, with portfolios poorly diversified across states and across the industries that drive state-level revenue (Babina, Jotikasthira, Lundblad, and Ramadorai, 2021).

Consistent with the pricing of idiosyncratic default risk, Schwert (2017) reports that municipal credit spreads correlate only weakly with aggregate credit-risk factors, and Wang, Wu, and Zhang (2008) attribute most of the cross-sectional variation in investment-grade municipal bond spreads to issuer-specific default and liquidity conditions. These findings suggest that a substantial share of the cross-sectional variation in municipal bond spreads reflects issuer-specific default conditions, even though common factors may also matter. To better isolate idiosyncratic factors, I use year fixed effects to absorb common time variation in municipal bond spreads and focus on cross-sectional differences across cities. As a further test of the distinction between systematic and idiosyncratic risk, in Section VI.A of the paper I show that the main results are robust to controlling for estimates of cities' systematic risk exposures.

II.B. A Distance-to-Default Framework

I formalize the determinants of the default component of the spread with a simple distance-to-default framework in the spirit of Merton (1974) and Leland (1994). City i 's fiscal capacity at horizon T , A_{iT} , is normally distributed with mean μ_{it} and variance σ_{it}^2 , and the city enters distress if fiscal capacity falls below a threshold B .¹ The spread, s_{it} is increasing in the market-implied probability of distress, which in turn depends on the city's distance to default measured in units of its own revenue volatility:

$$(1) \quad s_{it} = g(\Phi(-d_{it})), \quad d_{it} = \frac{\mu_{it} - B}{\sigma_{it}},$$

¹The endogenous determinants of this threshold—the elasticity of the tax base and the state-specific process for resolving financial distress—are modeled theoretically by Carlson, Giammarino, and Heinkel (2022).

where $\Phi(\cdot)$ is the standard normal CDF and $g(\cdot)$ is an increasing function that translates the distress probability into a yield premium.²

Two structural assumptions link the mean and variance of fiscal capacity to the city's observables. First, define the HHI of employment concentration as $h_{it} = \sum_{k=1}^K p_{kit}^2$, where p_{kit} is the share of total employment in sector k across the K industry sectors. Second, let C_{it} denote the city's *latent fiscal capacity* — the stock of revenue-raising potential available to the jurisdiction above and beyond the revenue currently collected. Latent fiscal capacity reflects the deep, slow-moving resources that a city can mobilize in the event of an adverse shock to its ongoing revenues, either through higher tax rates, alternative revenue instruments, or the liquidation of accumulated wealth. Section III.D develops the empirical proxy for C_{it} in detail and grounds it in the public-finance literature on fiscal capacity measurement.

With these definitions, the reduced-form assumptions are:

$$(2) \quad \mu_{it} = \mu_0 + \alpha C_{it} + \gamma h_{it},$$

$$(3) \quad \sigma_{it}^2 = \sigma_0^2 h_{it},$$

with $\alpha, \gamma, \sigma_0 > 0$. Equation (2) is linear for simplicity, but it generalizes to non-linear functions.

The key assumption in these two equations is that concentration has a positive relationship with both the mean and the variance of fiscal capacity. The variance equation (3) follows from a standard portfolio argument: if sector-level employment shocks have a common variance σ_0^2 and are uncorrelated after absorbing aggregate factors, the idiosyncratic variance of city revenue growth is exactly $\sigma_0^2 h_{it}$.³

The mean equation (2) captures the opposing force of concentration. A large urban-economics literature documents that industrial agglomeration raises local productivity through knowledge diffusion, labor-market pooling, and input sharing (Marshall, 1890; Jacobs, 1969; Glaeser, Kallal, Scheinkman, and Shleifer, 1992; Henderson, Kuncoro, and Turner, 1995; Duranton and Puga, 2004; Rosenthal and Strange, 2004; Ellison, Glaeser, and Kerr, 2010). Thus, cities with higher concentration could have larger tax bases with greater fiscal capacity.

Substituting Equations (2) and (3) into Equation (1), the city's distance to default becomes

$$(4) \quad d_{it} = \frac{\mu_0 - B + \alpha C_{it} + \gamma h_{it}}{\sigma_0 \sqrt{h_{it}}}.$$

²The monotone mapping g absorbs expected loss given default, bond maturity, the risk-free discount rate, and the price of risk. The comparative statics below depend only on $g' > 0$.

³I show below that residual cross-sector employment-growth correlations across the twenty NAICS supersectors are approximately zero once year fixed effects are removed.

Industrial concentration enters the numerator linearly through agglomeration and enters the denominator as $\sqrt{h_{it}}$ through the risk channel. The two effects pull in opposite directions, with the balance moderated by γ and σ_0 .

II.C. Comparative Statics and Hypotheses

Differentiating the spread s_{it} with respect to h_{it} yields

$$(5) \quad \frac{\partial s_{it}}{\partial h_{it}} \propto \phi(d_{it}) \cdot \frac{(\mu_0 + \alpha C_{it} - B) - \gamma h_{it}}{2 \sigma_0 h_{it}^{3/2}},$$

where ϕ is the standard normal density. The numerator separates the two channels of the HHI effect. The first term, $\mu_0 + \alpha C_{it} - B$, captures the risk channel. Whenever the city's mean fiscal capacity lies above the distress threshold, increasing concentration increases the spread. The second term, $-\gamma h_{it}$, captures the agglomeration channel: higher HHI raises the mean fiscal capacity and shrinks the default probability. The sign of the total effect depends on which channel is larger, and so it is an empirical question.

In contrast, differentiating the spread with respect to latent fiscal capacity,

$$(6) \quad \frac{\partial s_{it}}{\partial C_{it}} \propto -\phi(d_{it}) \cdot \frac{\alpha}{\sigma_0 \sqrt{h_{it}}} < 0,$$

yields an unambiguously negative relationship: latent fiscal capacity enters only through the mean and has no offsetting channel.

These comparative statics yield the first two hypotheses of the paper.

Hypothesis 1 (Industrial Concentration). *The effect of industrial concentration on municipal bond spreads is ambiguous in sign. Concentration raises spreads through the risk channel and lowers spreads through the agglomeration channel. Which force dominates on average is an empirical question.*

Hypothesis 2 (Latent Fiscal Capacity). *Greater latent fiscal capacity lowers municipal bond spreads.*

The more distinctive prediction of the framework concerns the interaction between h_{it} and C_{it} . Differentiating (5) once more with respect to C_{it} ,

$$(7) \quad \frac{\partial^2 s_{it}}{\partial h_{it} \partial C_{it}} \propto \frac{\alpha \phi(d_{it})}{2 \sigma_0 h_{it}^{3/2}} \cdot \left[1 - d_{it}^2 + \frac{2\gamma\sqrt{h_{it}}}{\sigma_0} d_{it} \right].$$

The sign of the cross partial is determined by the expression in brackets. In the absence of agglomeration ($\gamma = 0$), the bracket reduces to $1 - d_{it}^2$, which is negative whenever the city is more than one standard deviation from its distress threshold ($d_{it} > 1$). With $\gamma > 0$, the

bracket is negative for values of d_{it} above a slightly higher cutoff,

$$(8) \quad d^* \equiv \frac{\gamma\sqrt{h_{it}}}{\sigma_0} + \sqrt{\left(\frac{\gamma\sqrt{h_{it}}}{\sigma_0}\right)^2 + 1}.$$

For reasonable parameter values, d^* is modestly above 1, which is comfortably below the implied distances to default of most investment-grade issuers. The intuition is that with ample latent fiscal capacity, a city’s default threshold lies in the far tail of the default-probability distribution, where there is little sensitivity to HHI-induced variance. A city with less latent fiscal capacity has its default threshold closer to the center of the probability distribution, where the sensitivity is larger. For the sample studied in this paper — investment-grade issuers with implied distances to default well above d^* for reasonable parameter values — the framework predicts a negative cross partial.

Hypothesis 3 (Interaction). *For cities whose fiscal capacity is comfortably above the distress threshold, the effect of industrial concentration on bond spreads is larger in absolute value in cities with less latent fiscal capacity.*

This hypothesis follows from the portfolio-variance result in (3) combined with the unimodal shape of the fiscal-capacity distribution. The mechanism requires only that the marginal contribution of volatility to the default probability be largest near the distress threshold and decay in the tails, a property shared by any single-peaked distribution.

III. DATA AND SAMPLE CONSTRUCTION

I construct a city-year panel of municipal bond spreads and employment concentration by linking secondary-market bond trading records to Census employment data. This section describes the data sources, variable construction, and sample composition.

III.A. Municipal Bond Trading Data

Bond pricing data come from the Municipal Securities Rulemaking Board (MSRB) via its Electronic Municipal Market Access (EMMA) system, which records every municipal bond trade executed in the United States with the trade date, yield-to-maturity, par traded, and counterparty type. From the universe of secondary-market trades over 2005–2022, I apply six filters before computing spreads. First, I retain only customer-side trades (sales and purchases by customers, $S + P$) and drop interdealer trades, which represent inventory transfers between dealers rather than arm’s-length pricing (Harris and Piwowar, 2006; Schwert, 2017). Second, I drop trades on bonds with less than one year of remaining maturity,

where near-maturity pricing introduces noise. Third, I impose a 60-day post-issuance black-out: any trade within 60 days of the bond’s dated date is dropped, removing the period of price discovery and underwriter support that follows new issuance. Fourth, I drop trades on pre-refunded bonds—bonds whose payments are escrowed in Treasury or agency securities and whose yields therefore reflect the escrow’s collateral rather than the issuer’s credit. A bond is flagged as pre-refunded for a given trade when its yield falls at least 25 bp below an active duration-matched Treasury proxy and the share of such trades exceeds 1.5% of the bond’s lifetime trade history. Fifth, I drop trades on bonds priced below par by more than the IRS de-minimis threshold of 0.25% of face value times years to maturity, where market discount is taxed at ordinary income rates and pricing differentials are driven by the differential tax treatment rather than by credit. Sixth, I retain only city-years with at least three individual trades drawn from at least two distinct bonds, reducing noise from thinly traded issues. The final sample includes cities from 44 states plus Washington D.C.⁴

For each surviving trade I compute the AAA-cities-benchmark yield spread by subtracting a contemporaneous, maturity-matched benchmark yield from the trade’s yield-to-maturity:

$$(9) \quad s_{j,t} = [y_{j,t} - r_{m_j, \tau_j}^{\text{AAA}}] \times 100 \quad (\text{basis points}),$$

where $y_{j,t}$ is the yield-to-maturity of trade j executed on date t , τ_j is the bond’s remaining maturity, and $r_{m_j, \tau_j}^{\text{AAA}}$ is the AAA-cities-benchmark yield in calendar month m_j at the tenor closest to τ_j .

The AAA-cities benchmark is a monthly yield curve at seven tenor buckets (1, 2, 5, 7, 10, 20, and 30 years) constructed from trades issued by cities rated AAA by any of S&P, Moody’s (Aaa), Fitch, or Kroll in the trade’s year. For each (year-month, tenor) cell, the benchmark is the par-weighted mean yield-to-maturity of AAA-rated city trades whose remaining maturity falls in the tenor window. Cells lacking direct AAA trades are filled, in turn, by (i) forward-fill within the same tenor across up to three consecutive months; (ii) log-linear cross-tenor interpolation within the month using surrounding tenors; and (iii) the year-tenor cell from an annual AAA muni curve as a last resort. In the construction sample, all 1,512 (year-month, tenor) cells are populated by direct AAA-rated trades, so no fallbacks are triggered. Trades from non-AAA cities are excluded from the benchmark itself but their spreads are still computed against it, allowing the benchmark to serve as a clean, credit-quality-anchored reference yield curve while spreads of below-AAA cities measure their compensation for credit risk relative to that anchor.

⁴States with small populations tend to have thinly traded bonds. Hence, the sample does not include any cities in Alaska, Delaware, Hawaii, Idaho, North Dakota, or South Dakota.

Trade-level spreads are aggregated to par-weighted bond-month means; bond-month spreads are then aggregated to par-weighted city-year means, producing a single spread Spread_{ct} for each city c in each year t . All spread and yield variables are winsorized at the within-year 1st and 99th percentiles to mute single-trade outliers.

The results that follow are not sensitive to the choice of the AAA-cities benchmark. Internet Appendix Tables IA.6 and IA.12 replicate the OLS and IV specifications using the raw par-weighted yield with no benchmark subtraction at all, and the HHI gradient, the cross-tercile wedge, and the IV coefficients are quantitatively unchanged. In untabulated results, the findings are likewise unchanged when the dependent variable is replaced by a maturity-matched Treasury-benchmarked spread (with a constant tax adjustment). The AAA-cities benchmark is therefore a convenience that strips out common municipal-curve movements and is not driving the documented patterns.

III.B. Entity Matching

Linking MSRB bond records to Census economic data requires matching MSRB issuer identifiers to Census place or county subdivision codes. I construct this crosswalk in two steps. First, I parse MSRB security descriptions to extract the city name and state associated with each six-digit CUSIP issuer code. Second, I fuzzy-match extracted city names to a master entity concordance of Census places and county subdivisions using token-set-ratio string similarity, requiring a minimum match score of 85 on a 0–100 scale. I manually verify ambiguous matches and correct errors in the automated crosswalk.

For cities in Minor Civil Division (MCD) states—including Connecticut, Massachusetts, New York, New Jersey, Pennsylvania, Ohio, Michigan, and Illinois—local governments are organized as county subdivisions (towns and townships) rather than incorporated places. I use the ACS county subdivision geographic identifier for these entities, ensuring consistent matching across state governance structures.

III.C. Employment Concentration

An important explanatory variable of this paper is the industrial concentration of cities. Because my unit of analysis is a city, it is important that my measure of industrial concentration is at the city level. Using county-level sector distributions would generate imprecise measures of a city’s economic condition, especially for smaller cities in large counties. The most precise data available are employment data from the Workplace Area Characteristics (WAC) files of the U.S. Census Bureau’s Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES). LODES provides a

near-census of jobs covered by state unemployment insurance programs, geocoded to Census blocks and classified into 20 NAICS supersectors (CNS01 through CNS20 in the LEHD classification). I aggregate employment from block level to Census places using the Census Bureau’s geographic crosswalk, producing sector-level employment counts for each city and year from 2002 to 2022.⁵

The 20-sector NAICS supersector classification is coarser than finer industry classifications available from establishment-level datasets. The coarser aggregation is a feature of the research design rather than a compromise. First, the theoretical identification of HHI as the natural scaling of idiosyncratic fiscal variance requires approximately orthogonal sectoral shocks after absorbing aggregate factors; at fine resolution this fails because closely related sub-industries (e.g., logging and wood pulp) share input, technology, and demand exposures. Second, LODES is the only public dataset that reports employment at the Census place level, which is the level at which municipal debt is issued and repaid. Third, finer classifications suffer from disclosure suppression at the place-industry level, which artificially inflates measured HHI. I verify the first condition empirically below: across the 20 supersectors, the mean pairwise residual correlation after absorbing year fixed effects is approximately zero.

I use workplace-based employment (WAC) throughout the analysis because municipal debt is serviced by the taxing jurisdiction’s own revenue base. WAC captures economic activity within city boundaries—the employers whose presence generates commercial property value and tax revenue. When a plant closes, the host city’s tax base contracts, making workplace-based concentration the relevant measure of fiscal exposure.

I measure industrial concentration using the Herfindahl–Hirschman Index:

$$(10) \quad \text{HHI}_{ct} = \sum_{k=1}^K p_{kct}^2,$$

where $p_{kct} = E_{kct}/E_{ct}$ is sector k ’s share of total employment in city c in year t , and $K = 20$ is the number of NAICS supersectors. HHI ranges from $1/K = 0.05$ for a perfectly diversified city to 1 for a single-sector economy.

The distribution of sector employment shares within cities is right-skewed: most sectors account for small fractions of local employment, while a few dominant sectors hold disproportionately large shares. This pattern mirrors the well-documented skewness of firm size distributions (Gibrat, 1931; Gabaix, 1999) and establishment counts (Luttmer, 2007), extending those regularities to the within-city sector level. The right skew implies that HHI

⁵LODES covers approximately 95 percent of private employment through state unemployment insurance records, plus federal employment. Self-employment and informal employment are excluded.

is driven primarily by the one or two largest sectors in each city, making the concentration measure sensitive to shocks in dominant industries.

HHI is the natural measure of revenue-volatility risk because, under sector shocks with equal variance that are uncorrelated after absorbing aggregate factors such as year fixed effects, the variance of a city’s aggregate revenue is proportional to its HHI, as shown in Section II.B. I verify that this condition holds in my data. Unconditionally, average pairwise correlations of national sector employment growth across the 20 NAICS supersectors are positive and economically meaningful (mean $\rho \approx 0.26$), reflecting the shared influence of the aggregate business cycle. After controlling for year fixed effects, however, the average residual correlation collapses to essentially zero (mean $\rho \approx -0.01$). In other words, the common time component accounts for nearly all of the cross-sector co-movement in the data, and what is left after absorbing year fixed effects is plausibly idiosyncratic.⁶ This supports the use of HHI as a scaling factor for idiosyncratic revenue volatility in regressions that include year fixed effects and justifies my choice of HHI over an alternative portfolio-variance measure that would require estimating a full 20×20 sector covariance matrix at the city level, which is infeasible with available data.

III.D. Latent Fiscal Capacity

Hypothesis 2 states that spreads are decreasing in the city’s latent fiscal capacity C_{it} — the stock of revenue-raising potential available to the jurisdiction beyond the taxes it currently collects. This subsection defines the construct, grounds it in the public-finance literature, and defends the use of median property value as its empirical proxy.

The concept of latent fiscal capacity has a long tradition in public finance, originating with efforts to measure the revenue that a jurisdiction *could* raise under standardized tax effort rather than the revenue it currently collects. Ladd and Yinger (1989) define a city’s fiscal capacity as the size of its underlying tax bases and argue that the long-run solvency of a municipality is determined by the health of these bases rather than by its current revenue flow. A city with large tax bases has the option to raise revenue by increasing its tax effort when ongoing revenues come under stress, and this option is precisely the “cushion” that separates a city with ample latent capacity from one near distress.

I proxy for latent fiscal capacity with log median home value at the city level, computed from the American Community Survey.⁷ Median home values proxy for fiscal capacity

⁶Appendix Table IA.1 in the Internet Appendix reports the full unconditional and conditional correlation matrices.

⁷For the 2009–2022 period, I use ACS 5-year estimates for places and county subdivisions. For 2005–2008, I extend coverage using ACS 3-year estimates (available for places with population above 20,000), ACS 1-year

through two well-documented channels. First, property values directly determine municipal revenue capacity through multiple fiscal instruments: property tax receipts, property transfer taxes, and income and sales tax revenue via household wealth effects. Lutz, Molloy, and Shan (2011) document that house prices affect local government revenue through all of these channels in the cross-section of U.S. cities, and Mian and Sufi (2014) and Mian, Rao, and Sufi (2013) provide causal evidence that housing wealth drives consumption and employment, and Adelino, Schoar, and Severino (2015) document a causal effect of housing collateral on entrepreneurship; together these channels support the non-property components of a city’s tax base. Cities with higher median home values therefore have systematically broader and more resilient tax bases.

Second, property values proxy for latent capacity to raise alternative revenue during periods of fiscal stress. Even in jurisdictions with statutory property-tax limitations such as California’s Proposition 13, municipalities with higher property values retain access to special assessments, user fees, parcel taxes, and voter-approved bond measures, and wealthier residents have greater economic capacity to absorb higher burdens on any of these margins.

A related question is whether intergovernmental transfers should be treated as part of latent fiscal capacity. I do not include them as a separate regressor for two reasons. First, transfers are endogenous to local distress: Gao, Lee, and Murphy (2019) show that “proactive” states run counter-cyclical transfers to distressed municipalities, raising intergovernmental revenue as a share of local revenue in weak economic conditions. Therefore, conditioning on observed transfers would introduce a “bad controls” problem. As noted by Angrist and Pischke (2009), good controls are variables plausibly fixed before the regressor of interest is determined, while bad controls are variables that “might just as well be dependent variables too.” Second, the cross-sectional variation in transfer intensity is driven largely by state-level rules, which are absorbed by city fixed effects and by the state $\times$ year fixed effects used as a robustness check.

In the empirical analysis I use both log median home value directly and, for heterogeneity tests, a within-year tercile assignment that allows cities to move between terciles as their property values rise or fall relative to the cross-section.

estimates (places above 65,000), and, for remaining gaps, a county-level proxy based on the 2009 ACS 5-year county estimate.

III.E. Additional Controls

In all of the regressions, I control for city fixed effects to absorb city-level time-invariant characteristics that could affect municipal bond pricing, such as persistent government structures, legal statutes, and geographic-based traits. I also control for year fixed effects to absorb all macroeconomic conditions common to all cities that could affect the time-series of bond yields.

To control for time-varying factors, I include the county-level unemployment rate (Bureau of Labor Statistics, Local Area Unemployment Statistics), the state top marginal income tax rate (Tax Foundation), and log population (Census Bureau’s Population Estimates Program).

I also control for two distinct sets of bond characteristics, scraped from EMMA’s security-detail pages. The first set summarizes the city’s *outstanding* bond portfolio in each year: the par-weighted GO bond share, par-weighted callable share, and par-weighted average years to remaining maturity, computed across all of the city’s outstanding bonds (not just those traded in the year). These outstanding-portfolio characteristics describe the city’s issued bond structure—a city-level attribute that could itself reflect local fiscal conditions—and absorb the corresponding omitted-variable channel. The second set summarizes the *traded* bonds within each city-year that enter the spread computation: the par-weighted GO share, callable share, and average years to maturity of the traded bonds, plus the log number of trades and log par value traded. These traded-sample characteristics address selection bias in which specific bonds change hands in a given city-year—short-maturity GO bonds, for instance, may trade more frequently than long-maturity revenue bonds—and ensure that the HHI coefficient is not driven by composition shifts in the trading sample.⁸

As a robustness check, I augment the baseline specification with city-level demographic characteristics from the American Community Survey: racial composition (percent white, Black, and Hispanic), median age, percent foreign-born, and percent with a bachelor’s degree. This test is motivated by Butler and Yi (2022), who find that demographic traits predict municipal bond yields, raising the concern that HHI could proxy for demographic composition rather than capturing an independent economic channel. None of these demographic variables are individually significant predictors of spreads conditional on the other controls ($p > 0.25$ for all), and the HHI coefficient and its interaction with property value terciles are nearly identical with and without demographics. Industrial concentration already subsumes

⁸Appendix C reports further robustness tests using four additional trade-level liquidity measures—bid-ask proxy, yield dispersion, zero-trading-day fraction, and turnover—constructed from the underlying MSRB transactions. The results are robust to all specifications.

whatever variation demographics would explain, confirming that HHI captures the relevant underlying economic channel.

Fiscal-outcome variables, including leverage, credit ratings, and reserve-fund balances, are intentionally excluded from the baseline specification. The reason is not that these variables are unimportant for municipal credit risk, but like intergovernmental transfers, they are bad controls: they are themselves outcomes of the same local economic environment whose pricing implications I seek to estimate. City fixed effects further reduce the need to control for these fiscal balance-sheet variables because much of the cross-sectional variation in fiscal primitives is persistent and shaped by institutional constraints, including state-imposed debt limits and balanced-budget rules (Kiewiet and Szakaly, 1996; Epple and Spatt, 1986). Appendix B confirms this empirically: leverage ratios and intergovernmental revenue shares are nearly constant across property-value terciles, while bond spreads vary substantially over the same partition.

As a robustness check, I estimate a specification that includes the long-term debt-to-revenue ratio, together with a missing-data indicator to preserve the main sample. The HHI coefficient and its interaction with property values change by less than one percent of a standard error, and the leverage coefficient itself is statistically and economically zero. This pattern is consistent with the fixed effects absorbing the persistent cross-sectional component of leverage, and with contemporaneous fiscal outcomes being inappropriate baseline controls rather than omitted predetermined covariates.

III.F. Sample Characteristics

The final analysis panel contains 21,927 city-year observations spanning 1,177 cities from 2005 to 2022. The panel is unbalanced: not all cities have sufficient secondary-market trading activity in every year. Of these, 21,701 observations (99 percent) have non-missing employment concentration data from LODES; the remainder are primarily county or special-district issuers for which Census place-level employment data are unavailable. Of the 21,701 observations with HHI data, 18,447 have non-missing values for all regression controls. Of these, 18,427 remain after dropping singleton city-year groups in the fixed effects estimation and constitute the estimation sample reported in the OLS tables.

Figure I maps the sample cities, with bubble size proportional to average annual par value traded. Sample cities cluster around large metropolitan areas, though many smaller cities in more rural areas also appear. Trading activity is highly skewed: a small number of large cities account for the bulk of secondary-market par volume, while many smaller cities trade infrequently. The requirement of at least three secondary-market trades per city-year tilts

the sample toward larger and more actively traded cities. Because the interaction results are concentrated in low-property-value cities—which are, on average, smaller—this selection works *against* finding the reported effect: the population most sensitive to concentration is also the population most likely to be excluded.

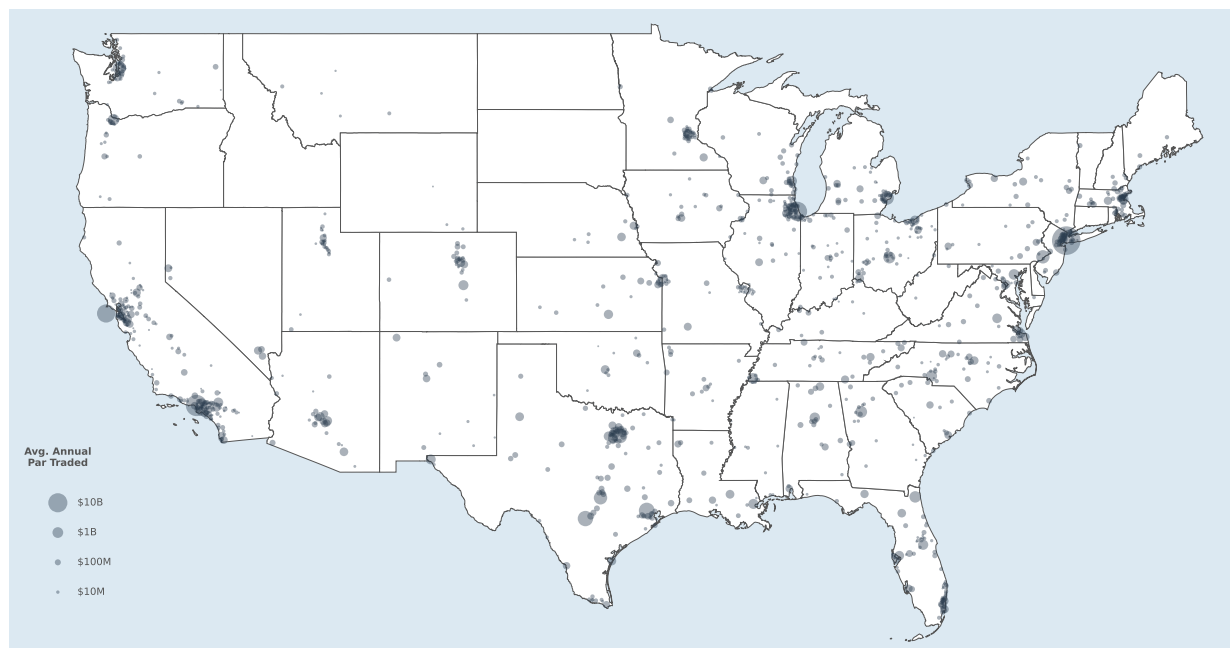


FIGURE I
Geographic Distribution of Sample Cities and Trading Volume

Notes: Each bubble represents a city in the analysis sample. Bubble area is proportional to the city's average annual par value of municipal bonds traded in the secondary market over the 2005–2022 sample period. Larger cities are plotted on top of smaller cities so that metropolitan centers remain visible. The sample spans 1,177 cities across 44 states and Washington D.C.

Table I reports summary statistics for the 18,447 city-year observations used in the OLS regressions. The mean AAA-cities-benchmark spread is approximately 21 basis points with a standard deviation of 50 basis points; the median is 11 basis points and the interquartile range spans -4 to 33 basis points. The negative lower quantile reflects that 86 percent of sample bonds carry an AAA or AA rating, so a meaningful fraction of trades price inside the AAA-cities benchmark curve. The mean HHI of 0.120 corresponds to an effective number of equal-sized sectors of approximately 8.3, indicating moderate concentration relative to the 20-sector maximum. The median city has a population of approximately 56,000.

These summary statistics are broadly consistent with prior municipal bond studies. Schwert (2017) reports a mean tax-adjusted spread of approximately 106 basis points and a mean remaining maturity of 13 years in a sample of primary-market offerings; his benchmark is the

Treasury curve, so the level is not directly comparable to the within-AAA spread reported here, which by construction strips out the AAA-versus-Treasury wedge that dominates the level of any Treasury-benchmarked muni spread. Gao, Lee, and Murphy (2020) report a mean GO share of approximately 50 percent and a median population of roughly 50,000 in their city-level sample, comparable to the GO share and city size distributions in this sample. These statistics indicate the sample is a good representation of the municipal bond market.

TABLE I
Summary Statistics

	Mean	Std. Dev.	P25	Median	P75
Yield spread (bps)	21.100	50.100	−3.700	11.300	33.300
HHI (employment concentration)	0.120	0.039	0.098	0.111	0.130
Median property value (\$000s)	257.100	199.700	132.100	196.800	309.800
Population (000s)	116.900	337.700	36.600	56.000	98.200
State top income tax rate (%)	5.650	3.700	3.400	5.800	8.820
Unemployment rate (%)	6.000	2.500	4.100	5.300	7.500
GO share (traded)	0.476	0.419	0.001	0.425	0.972
Callable share (traded)	0.320	0.296	0.058	0.242	0.512
Avg. years to maturity (traded)	9.700	4.700	6.400	9.000	12.100
Par value traded (\$M)	114.100	1136.700	3.700	12.300	37.800
Trades per city-year	595	4,538	47	132	363
GO share (outstanding)	0.421	0.361	0.063	0.342	0.770
Callable share (outstanding)	0.361	0.214	0.198	0.341	0.503
Avg. years to maturity (outstanding)	9.000	3.700	6.400	8.500	11.100

Notes: Sample restricted to city-year observations with non-missing AAA-cities-benchmark spread, HHI, and all regression controls (18447 city-years, 1177 unique cities). The dependent variable is each trade's yield minus the par-weighted yield of AAA-rated cities' GO trades in the same year and maturity bucket, then par-weighted to the city-year level (in basis points). Both callable and non-callable bonds are included. HHI is the Herfindahl–Hirschman Index of employment concentration across 20 NAICS supersectors from LODES. Median property value is from the American Community Survey 5-year estimates. Traded bond characteristics are par-weighted across each city-year's traded bonds; outstanding characteristics are par-weighted across each city-year's outstanding bonds. All spread and yield variables are winsorized at the within-year 1st and 99th percentiles.

In unreported tests, I find that larger cities are more diversified, as expected. In particular, industrial concentration is negatively correlated with log population ($\rho \approx -0.30$). I also find that concentration is negatively correlated with log median property value ($\rho \approx -0.15$): less affluent cities tend to be more concentrated.

IV. REDUCED-FORM EMPIRICAL EVIDENCE

This section documents that employment concentration predicts higher municipal bond spreads, but that this relationship is concentrated among cities with limited fiscal capacity. I first show univariate evidence that yield spreads are higher for concentrated cities, particularly among cities with low property values, and confirm in Appendix D that the cross-sectional relationship holds in a pure between-city regression with state fixed effects. I then estimate panel OLS regressions with city and year fixed effects and a comprehensive set of controls that isolate within-city variation, progressively tightening the identification to move toward a causal interpretation.

IV.A. Univariate Evidence

Table II provides univariate evidence on yield spreads by industrial concentration and property values. Among cities in the lowest tercile of property values, those above the median of industrial concentration have AAA-cities-benchmark spreads of 31.2 bps compared to 20.4 bps for cities below the median, a statistically significant difference of 10.8 bps ($t = 3.5$). In contrast, among cities with the highest property values, the spreads are statistically indistinguishable between cities with high concentration (28.8 bps) and low concentration (23.8 bps), a difference of 5.0 bps ($t = 1.2$). These results provide the first evidence that industrial concentration is correlated with higher costs of debt, but only when the latent fiscal capacity of the city is low.

IV.B. Multivariate Fixed Effects Regressions

To isolate the effect of employment concentration from potential confounders, Table III reports fixed effects panel OLS estimates. Column (1) shows that after including city and year fixed effects, the unconditional HHI coefficient is positive but not statistically significant at conventional levels. Column (2) shows that property values are statistically negatively correlated with bond spreads, as expected. Columns (3)–(5) interact HHI with the three property-value terciles. Because high-property-value cities have the lowest mean spreads and serve as the natural benchmark, I report results with the high tercile as the omitted reference category, so the HHI base coefficient is the HHI gradient for high-property-value cities and the two interaction rows report deviations of the lower terciles from that benchmark.

The central test of Hypothesis 3 is whether the HHI gradient *differs* across property-value terciles — not whether any individual tercile’s gradient differs from zero. The data answer this clearly. In Column (3) the $\text{HHI} \times \text{Low PV}$ interaction is +189.4 basis points per unit

TABLE II
Mean Bond Spreads by Industrial Concentration and Property Value

	Low Property Value	High Property Value	Difference (High – Low)
High Concentration	31.15	28.82	-2.34 (-0.61)
Low Concentration	20.39	23.82	3.43 (1.02)
Difference (High – Low)	10.76*** (3.51)	4.99 (1.23)	-5.77 (-1.13)

Notes: Entries are mean AAA-cities-benchmark municipal bond yield spreads in basis points, averaged at the city level over the sample period. Cities are split at the sample median HHI (High/Low Concentration) and into property-value terciles (HP1/HP3); the middle tercile (HP2) is omitted. Cell counts: High Conc/Low HP = 225, High Conc/High HP = 172, Low Conc/Low HP = 168, Low Conc/High HP = 220. Robust *t*-statistics in parentheses. Difference-in-differences estimated by OLS with robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

HHI ($p < 0.01$): low-property-value cities exhibit an HHI–spread sensitivity that is 189 basis points *higher* than the sensitivity in high-property-value cities. This wedge is robust across specifications, attenuating modestly to 144.9 with macro controls (Column 4) and 126.1 with the full set of macro and bond controls (Column 5), but remaining significant at the 5 percent level in both. The wedge between mid- and high-PV cities (HHI $\times$ Mid PV) is also positive in all three columns and significant in Column (3). The two interaction coefficients exhibit a monotonic pattern: the gap between the low and high terciles is roughly twice the gap between the mid and high terciles, exactly the ordering predicted by the framework’s cross partial.⁹

The point estimates of the per-tercile gradients have intuitive signs — approximately +109 basis points per unit HHI in low-PV cities, near zero in the middle, and –81 in high-PV cities (Column 3) — but these individual gradients are estimated less precisely than the cross-tercile wedge because they involve a sum of two coefficients each. The economically and statistically robust statement is that two cities with identical HHI changes face *opposite-signed* spread responses depending on their property-value position, with the implied 189-basis-point gap being significant well below conventional thresholds. In economic units, the cross-tercile wedge translates into a $189 \times 0.039 \approx 7.4$ basis point spread differential between

⁹The same wedge appears under a coarser high-versus-low binning at the within-year property-value median: the HHI $\times$ Low PV interaction is +133 basis points without controls ($p < 0.01$), +106 with macro controls ($p < 0.05$), and +87 with the full controls ($p < 0.10$), about 70 percent of the tercile-spec magnitudes as expected when comparing halves rather than extreme thirds.

low- and high-PV cities for a one-standard-deviation HHI move—about a third of the 21 basis point sample mean spread.

TABLE III
Industrial Concentration and Municipal Bond Spreads: OLS Evidence

	(1)	(2)	(3)	(4)	(5)
HHI	13.45 (29.00)		−80.83* (41.11)	−45.41 (37.51)	−39.25 (37.41)
ln(Property value)		−23.92*** (6.44)			
Low Property Value			−18.01 (10.40)	−12.75 (8.48)	−10.88 (8.51)
Mid Property Value			−3.68 (7.06)	−1.67 (5.81)	−0.68 (5.76)
Low Property Value × HHI			189.44*** (58.20)	144.85** (56.56)	126.10** (57.35)
Mid Property Value × HHI			79.99* (44.89)	56.44 (40.64)	42.81 (39.83)
Macroeconomic controls	No	No	No	Yes	Yes
Bond controls	No	No	No	No	Yes
City and year fixed effects	Yes	Yes	Yes	Yes	Yes
Observations	18,745	21,701	18,741	18,427	18,427
Adjusted R^2	0.365	0.369	0.366	0.386	0.415

Notes: The dependent variable is the AAA-cities-benchmark municipal bond yield spread, in basis points. HHI is the Herfindahl index of city-level employment shares across 20 industry sectors. Property-value terciles are formed within each year from ACS median home value; the high tercile is the omitted reference category, so the HHI base coefficient in Columns (3)–(5) is the HHI gradient for high-property-value cities, and the “Low × HHI” and “Mid × HHI” rows report how the gradient differs in lower terciles relative to high. All specifications include city and year fixed effects. Standard errors (in parentheses) are two-way clustered by city and year. Column (1) reports the unconditional HHI coefficient. Column (2) reports the unconditional log median property value coefficient. Columns (3)–(5) interact HHI with the property-value terciles. “Macro” controls are the state income tax rate, county unemployment rate, and log population. “Bond” controls are the GO share, callable share, and average years to maturity of outstanding and traded bonds, plus log number of trades and log par value traded. Coefficient estimates for the controls are reported in Internet Appendix Table IA.5. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

To contextualize the magnitudes, consider a low-property-value city at the 75th percentile of HHI (0.130) relative to the 25th percentile (0.098). The implied low-PV gradient of 108.6 basis points per unit HHI (recovered from Column (3) as $-80.8 + 189.4$) implies a spread difference of $108.6 \times (0.130 - 0.098) \approx 3.5$ basis points for this interquartile move, roughly 17% of the low-PV mean spread.

This magnitude is comparable to estimates from prior research. Gao, Lee, and Murphy (2020) find that newspaper closures—which reduce external monitoring of local government finances—raise municipal bond offering yields by 5–11 basis points. The one-standard-deviation effect of 4.2 basis points for low-property-value cities is of the same order of magnitude as the newspaper-closure effect. The HHI effect is also economically meaningful relative to other determinants of municipal spreads. A one-percentage-point increase in the county unemployment rate is associated with a 5.8 basis point increase in spreads (Internet Appendix Table IA.5, Column 5), comparable in magnitude to the 4.2 basis point effect of a one-standard-deviation increase in HHI among low-property-value cities.

Internet Appendix F formalizes this comparison with a Shapley decomposition of spread variation across four economic channels (the HHI channel, local macro and demographic controls, state taxation, and liquidity) after partialing out city and year fixed effects and bond-mix composition characteristics. The concentration channel accounts for approximately fourteen percent of the economic spread variation—nontrivial and of comparable magnitude to the canonical drivers identified in prior work.

Internet Appendix Table IA.10 shows that these results strengthen after the inclusion of state $\times$ year fixed effects, which absorb all time-varying state-level policies including changes to fiscal oversight, tax regimes, and macroeconomic conditions. Under this more demanding specification with the full set of macro and bond controls (Column 5), the implied low-property-value HHI gradient increases from 86.9 to 113.6 basis points (about a 30 percent strengthening), and the cross-tercile wedge—the HHI $\times$ Low PV interaction—is essentially unchanged at 125.5 basis points ($p < 0.05$, vs. 126.1 in the main specification). The fact that the HHI gradient and its interaction with property values are stable or stronger when state-level confounders are absorbed provides additional confidence that the HHI–spread relationship reflects city-level industrial structure rather than state-level factors correlated with concentration.

In addition, Internet Appendix Table IA.4 shows that the same relationships exist in pure cross-sectional tests. Collapsing the panel to city-level means and regressing average spreads on average HHI with state fixed effects, I find that the HHI gradient for low-property-value cities is 130 basis points ($p < 0.01$), declining to 83 for middle-tercile cities ($p < 0.05$), and 57 for high-value cities ($p < 0.10$). All three gradients are positive (unlike the within-city panel, where the high-tercile gradient is negative), but the monotonic decline with property value is preserved. This pure cross-sectional evidence establishes that the concentration–spread relationship is not an artifact of panel structure or within-city sample composition variation.

The results are robust to alternative dependent-variable definitions: Internet Appendix Tables IA.6, IA.7, and IA.8 reproduce the OLS specification using, respectively, the raw par-weighted yield (no AAA-cities-benchmark subtraction), the AAA spread restricted to actively-traded city-years (at least ten secondary-market trades), and the AAA spread under a loosened maturity filter (years to maturity in $[2, 25]$). The cross-tercile wedge and the low-property-value HHI gradient retain their sign, magnitude, and statistical significance across all three alternatives. Internet Appendix K further shows that the property-value-tercile interaction pattern survives when terciles are formed from the *prior year's* median property value, which is predetermined relative to current-year shocks and therefore unaffected by reverse causation. Internet Appendix L replaces both the dependent variable and HHI with their five-year backward moving averages—treating both objects as structural rather than year-by-year realizations—and finds that the cross-tercile wedge essentially doubles (from 126 to 266 basis points, $p = 0.022$) and the adjusted R^2 rises from 0.42 to 0.67, consistent with the year-by-year specification understating the structural relationship by attenuating the long-run signal with high-frequency noise.

V. CAUSAL EVIDENCE

The OLS results establish that employment concentration predicts higher municipal bond spreads, particularly among cities with limited fiscal capacity. However, the OLS gradient may not identify a causal effect if time-varying omitted variables drive both HHI changes and spread movements. Two endogeneity concerns are especially pressing. First, a city-specific economic shock—such as a plant closure or industry decline—could simultaneously concentrate employment (raising HHI) and deteriorate the fiscal outlook (raising spreads), so that HHI is a symptom of the shock rather than a causal driver of credit risk. Second, property values are endogenous responses to local economic conditions and may themselves respond to the same concentration shocks. Because HHI and property values are jointly determined by local conditions, neither can be treated as exogenous, and both must be instrumented simultaneously.

V.A. Identification Strategy

To establish causality for both the main effects and their interaction, I estimate a three-endogenous-variable instrumental variables design. The second-stage equation is

$$(11) \quad s_{ct} = \beta_1 \widehat{\text{HHI}}_{ct} + \beta_2 \widehat{\ln(\text{PV})}_{ct} + \beta_3 (\widehat{\text{HHI} \times \ln \text{PV}})_{ct} + \mathbf{X}'_{ct} \boldsymbol{\gamma} + \alpha_c + \delta_t + \varepsilon_{ct},$$

where $\widehat{\text{HHI}}_{ct}$ is city c 's instrumented employment HHI in year t , $\widehat{\ln(\text{PV})}_{ct}$ is the instrumented log of median property value, $(\widehat{\text{HHI} \times \ln \text{PV}})_{ct}$ is the instrumented interaction, $\mathbf{X}_{ct}$ is a vector of controls, and α_c and δ_t are city and year fixed effects. The IV specification uses the continuous interaction rather than the property-value terciles used in the OLS analysis because instrumenting for a discrete property-value classification is challenging in a model with city fixed effects; the within-city variation in group membership is too limited to identify meaningful variation.

To instrument for HHI, log property values, and their interaction, I use a set of excluded instrumental variables (IVs) constructed from two base instruments: a Bartik shift-share measure for concentration and a cumulative housing supply shock based on Guren, McKay, Nakamura, and Steinsson (2021) (GMNS). The IVs primarily target different endogenous variables: the Bartik IV predicts HHI while the GMNS IV predicts property values. The squares and cross-products of the IVs provide the additional variation needed to identify the interaction term.

V.B. Bartik-Style Instrument for HHI

I construct a shift-share instrument for employment concentration following the Bartik (1991) framework, building on the analysis of regional labor demand shocks developed by Blanchard and Katz (1992). The instrument combines predetermined city-level employment shares with exogenous national-level industry shocks to generate predicted variation in each city's industrial concentration.

Two recent frameworks clarify the identifying assumptions in shift-share designs. Goldsmith-Pinkham, Sorkin, and Swift (2020) show that shift-share instruments can be interpreted as exposure-weighted averages of sector-specific instruments, with identification requiring exogeneity of the base-period exposure shares. Borusyak, Hull, and Jaravel (2022) provide an alternative framework that shifts the identification burden to the shocks: validity requires that the industry-level shocks are as-good-as-randomly assigned conditional on controls, while the exposure shares serve only as weights. I follow the Borusyak, Hull, and Jaravel (2022) framework, using leave-one-out national industry employment growth as the shock source.

For each industry k , city c , and year t , the shock is

$$(12) \quad \tilde{g}_{kct} = [\ln(N_{kt} - E_{kc,t}) - \ln(N_{k,0} - E_{kc,0})] - [\ln(\bar{N}_t) - \ln(\bar{N}_0)],$$

where N_{kt} is national employment in industry k , $E_{kc,t}$ is city c 's own employment in sector k (excluded to avoid mechanical correlation), and the subtraction of aggregate employment

growth removes the common component absorbed by year fixed effects, leaving only relative cross-industry growth.

Because HHI is a nonlinear function of employment shares—specifically, the sum of squared shares—the instrument cannot be constructed as a standard linear Bartik. Instead, I propagate the national industry shocks through predetermined city employment to construct predicted sectoral employment:

$$(13) \quad \hat{E}_{kct} = E_{kc,0} \times \exp(\tilde{g}_{kct}),$$

where $E_{kc,0}$ is city c 's employment in sector k in the base year. The predicted concentration measure is then

$$(14) \quad \widehat{\text{HHI}}_{ct} = \sum_{k=1}^{20} \hat{s}_{kct}^2, \quad \text{where} \quad \hat{s}_{kct} = \frac{\hat{E}_{kct}}{\sum_{\ell} \hat{E}_{\ell ct}}.$$

This is analogous in spirit to the shift-share designs used by Autor, Dorn, and Hanson (2013) and others, but the aggregation is nonlinear because the target variable is concentration rather than employment growth. Under the exogenous-shares framework of Goldsmith-Pinkham, Sorkin, and Swift (2020), any function of exogenous base-year shares paired with the same shocks is itself a valid shift-share instrument. The complementary framework of Borusyak, Hull, and Jaravel (2025) delivers the same conclusion through exogeneity of the national sector shocks rather than of the shares: under shock exogeneity, any nonlinear shift-share construction built from those shocks inherits validity. The predicted-HHI instrument is therefore valid under the same identifying assumptions that validate a linear Bartik built from the same shares and shocks.

The identifying assumption is that national industry employment growth, after removing the aggregate component (absorbed by year fixed effects) and the city's own contribution (leave-one-out), is orthogonal to city-level unobservables that affect bond spreads, conditional on city fixed effects and controls. This assumption is plausible: conditional on city fixed effects, year fixed effects, and local economic conditions (unemployment, population, state tax rates), the remaining variation in the Bartik instrument comes from a city's differential exposure to national sector-specific trends—variation driven by the historical composition of industries, not by concurrent local economic shocks. The base-year employment data come from the 2002 LODES data, three years before the start of the bond-spread sample in 2005. The predetermined shares are fixed before the sample begins and therefore cannot be driven by contemporaneous credit deterioration or fiscal distress.

V.C. Housing Cycle Instrument for Property Value

Current property values are endogenous to local economic conditions: a negative employment shock that concentrates the industrial base also depresses local housing demand and prices. Simply controlling for property values in the spread regression would not resolve this simultaneity, and instrumenting HHI alone while treating property values as exogenous would bias the concentration estimate if unobserved local shocks affect both HHI and property values.

I construct a housing instrument following the approach developed by Guren, McKay, Nakamura, and Steinsson (2021), who estimate city-specific sensitivities to regional house-price cycles. The instrument exploits the fact that cities within the same Census region are exposed to common regional housing market forces, but respond with different elasticities due to differences in housing supply constraints, land-use regulation, and geographic features. For city c in Census region r , the annual GMNS shock is

$$(15) \quad z_{ct}^{\text{GMNS}} = \hat{\gamma}_c \times \Delta \ln(\text{HPI}_{rt}),$$

where $\hat{\gamma}_c$ is city c 's estimated sensitivity to regional house-price movements and $\Delta \ln(\text{HPI}_{rt})$ is the log change in the regional (Census division) house-price index. The sensitivity $\hat{\gamma}_c$ is estimated from a first-stage regression of city-level house-price growth on regional house-price growth, using data from the FHFA house price index data over the pre-sample period (1975–2004), requiring at least eight annual observations per CBSA. Cities with inelastic housing supply (e.g., geographically constrained coastal cities) have larger $\hat{\gamma}_c$, meaning that a given regional house-price shock translates into a larger local price change (Guren, McKay, Nakamura, and Steinsson, 2021).

The annual GMNS shock is naturally a flow instrument: it predicts house-price *growth* in a given year. However, the second-stage specification uses log property values in *levels*, not changes. Using a flow instrument for a stock variable creates a mismatch: a shock to house-price growth in year t affects the level of property values in all subsequent years, and a single year's flow carries limited information about the cumulative stock position. I therefore construct a cumulative instrument that aggregates the annual shocks over time:

$$(16) \quad Z_{ct}^{\text{GMNS}} = \hat{\gamma}_c \times \sum_{\tau=1}^t \Delta \ln(\text{HPI}_{r\tau}),$$

where $\sum_{\tau=1}^t \Delta \ln(\text{HPI}_{r\tau})$ is the cumulative log change in the regional house-price index from the base year through year t . Note the distinction from equation (15): the annual instrument z_{ct}^{GMNS} uses the single-period change $\Delta \ln(\text{HPI}_{rt})$, while the cumulative instrument Z_{ct}^{GMNS}

uses the accumulated level $\ln(\text{HPI}_{rt})$, capturing the total stock of house-price movements experienced by the city up to year t .

V.D. Identification of the Interaction

Identifying the interaction between HHI and property values requires instruments that provide variation in the product $\text{HHI} \times \ln(\text{PV})$ beyond what the two base instruments generate for the main effects alone. I construct additional instruments from squares and cross-products of the two base instruments: the Bartik predicted HHI (B) and the GMNS cumulative housing shock (G). Any nonlinear transformation of a valid instrument is itself a valid instrument, and the use of polynomial functions of base instruments to identify interaction effects is well established in the IV literature (Angrist and Pischke, 2009; Wooldridge, 2010). The squared terms B^2 and G^2 capture nonlinear responses of the endogenous variables to the instruments, while the cross-products BG , B^2G , and BG^2 generate variation in the joint distribution of concentration and property values. Intuitively, the cross-products are informative about the interaction because they capture cities where both concentration shocks (through B) and housing shocks (through G) are simultaneously large or small—precisely the variation needed to identify how the two channels interact.

The full instrument set $\{B, G, B^2, G^2, BG, B^2G, BG^2\}$ yields seven excluded instruments for three endogenous variables (HHI, $\ln(\text{PV})$, and $\text{HHI} \times \ln(\text{PV})$), providing four overidentifying restrictions that can be tested via the Hansen J statistic.

V.E. Main IV Results

Table IV presents the 2SLS estimates for the key variables. Internet Appendix Table IA.11 reports the complete results for all specifications, and Internet Appendix Tables IA.12, IA.13, and IA.14 replicate the IV estimates using the raw par-weighted yield, the actively-traded city-year subsample, and the loosened-maturity sample, respectively; the HHI gradient, property-value coefficient, and interaction term retain their signs and statistical significance across all three alternative dependent variables. All three columns instrument HHI, log property values, and their interaction simultaneously. Column (1) includes only city and year fixed effects; column (2) adds macro controls (unemployment rate, log population, state income tax rate); and column (3) further adds the full set of bond characteristics from the OLS specifications—both the outstanding-portfolio characteristics (par-weighted GO share, callable share, and average remaining maturity across the city’s full outstanding portfolio) and the traded-sample characteristics (par-weighted GO share, callable share, and average

maturity of the traded bonds, plus the log number of trades and log par value traded). Coefficients are scaled by 10^{-2} throughout the table; multiplying by 100 recovers basis points.

TABLE IV
2SLS Estimates of Concentration and Property Values on Bond Spreads

	(1)	(2)	(3)
Industrial Concentration (HHI)	121.855**	74.307**	66.486**
Standard error	(50.900)	(31.551)	(29.735)
<i>p</i> -value	0.017	0.019	0.025
Bootstrap <i>p</i>	0.020	0.014	0.022
ln(Property value)	−2.038***	−1.155***	−1.180***
Standard error	(0.624)	(0.374)	(0.354)
<i>p</i> -value	0.001	0.002	<0.001
Bootstrap <i>p</i>	0.006	0.003	0.001
HHI × ln(Property value)	−9.524**	−5.692**	−5.035**
Standard error	(4.264)	(2.618)	(2.472)
<i>p</i> -value	0.026	0.030	0.042
Bootstrap <i>p</i>	0.033	0.031	0.042
Macroeconomic controls	No	Yes	Yes
Bond controls	No	No	Yes
City and year fixed effects	Yes	Yes	Yes
Kleibergen-Paap <i>F</i>	10.541	11.243	11.361
Anderson-Rubin <i>p</i>	<0.001	<0.001	<0.001
CLR <i>p</i>	<0.001	<0.001	<0.001
Hansen <i>J p</i>	0.237	0.143	0.110
Observations	15,981	15,981	15,981

Notes: The dependent variable is the AAA-cities-benchmark municipal bond yield spread, in basis points (each trade’s yield minus the par-weighted yield of AAA-rated cities’ GO trades in the same year and maturity bucket, then par-weighted to the city-year level). All coefficients are scaled by 10^{-2} (i.e., units are percentage points; multiply by 100 for basis points). “ln(Property value)” is the log of ACS median home value. All three endogenous variables (HHI, ln(Property value), and their interaction) are instrumented by the seven instruments B , G , B^2 , G^2 , BG , B^2G , BG^2 , where B is predicted HHI from a Bartik shift-share design and G is the Guren et al. (2021) cumulative housing supply instrument. All specifications include city and year fixed effects (absorbed by pre-demeaning) and cluster standard errors by city. “Macroeconomic” controls are the state income tax rate, unemployment rate, and log population. “Bond” controls are the GO share, callable share, and average years to maturity of outstanding and traded bonds, plus log number of trades and log par value traded. Bootstrap *p*-values use the wild cluster bootstrap (999 replications, clustered by year). Anderson-Rubin (AR) and conditional likelihood ratio (CLR) tests are robust to weak instruments. Hansen *J* tests overidentifying restrictions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Consistent with Hypotheses 1–3, the IV coefficients have the predicted signs in every specification. HHI raises spreads (column 3 point estimate 66.5, $p < 0.05$), log property values

lower spreads (-1.18 , $p < 0.001$), and the interaction is negative and significant (-5.04 , $p < 0.05$). The negative interaction confirms that the concentration effect on spreads is larger for cities with lower property values. The Hansen J test does not reject overidentification in any specification ($p = 0.11$ – 0.24), and the Anderson–Rubin and conditional likelihood ratio tests strongly reject the null of zero coefficients ($p < 0.001$ in all columns), providing weak-instrument-robust confirmation that the instruments identify a real effect.

Economically, at the mean of log property values ($\overline{\ln(PV)} = 12.24$), a one-standard-deviation increase in HHI (0.039) raises spreads by approximately 19 basis points (column 3). This effect is substantially larger in low-property-value markets: at the 25th percentile of property values, the same HHI increase raises spreads by approximately 28 basis points, while at the 75th percentile the effect falls to roughly 11 basis points. The heterogeneity is consistent with the fiscal buffer mechanism: property values provide a cushion against fiscal stress, and this cushion attenuates the sensitivity of borrowing costs to industrial concentration.

All specifications cluster standard errors by city, which allows for arbitrary serial correlation within each city’s time series. Year fixed effects absorb common time shocks, so city clustering is the natural choice. I also report wild cluster bootstrap p -values (999 replications, clustered by year) throughout the table; these provide valid inference even with a small number of year clusters by resampling the cluster-level residuals rather than relying on asymptotic approximations. In the preferred specification (column 3), the bootstrap p -value for the interaction is 0.042 , confirming the standard inference.

An additional concern with any IV design is that weak instruments can distort standard Wald-based inference, leading to false rejections even when first-stage diagnostics appear adequate. The Anderson–Rubin (AR) test and the conditional likelihood ratio (CLR) test provide inference that is valid regardless of instrument strength. These tests have correct size even when the Kleibergen–Paap F -statistic is low. In the preferred specification, both the AR and CLR tests strongly reject the null that all endogenous coefficients are zero ($p < 0.001$), providing a robust confirmation that the instruments identify a real effect.

V.F. Diagnostic Evidence

Internet Appendix Table IA.15 reports the first-stage regressions for the preferred specification with all controls. The first stages show sensible patterns: the Bartik instrument (B) primarily predicts concentration, entering strongly in the HHI first stage (coefficient 0.84 , $t = 2.71$) and in the interaction first stage (coefficient 10.62 , $t = 2.93$), while the GMNS housing supply shock (G) primarily predicts property values (coefficient 0.55 , $t = 5.24$).

The cross-product instruments (BG , B^2G , BG^2) provide the additional variation needed to identify the interaction term, with several entering significantly across the three first stages. First-stage diagnostics are reassuring: the Kleibergen–Paap F -statistic is 11.4, exceeding the Stock and Yogo (2005) critical value of 8.5 for 10% maximal IV relative bias with three endogenous variables. Population negatively predicts concentration, consistent with larger cities having more competitive, diversified labor markets. The R^2 values including city and year fixed effects are 0.886, 0.964, and 0.883 for HHI, log property values, and the interaction, respectively, indicating that the fixed effects and instruments jointly explain the vast majority of variation in the endogenous variables.

Internet Appendix I reports two placebo Bartik tests that directly examine the exclusion restriction. The first randomly permutes the observed national sector shocks across sectors, destroying the sector-specific identifying variation while preserving the shocks’ magnitude and within-year dispersion. The second replaces contemporaneous shocks with future (year $t + 5$) shocks, isolating any persistent correlation between long-run sector trajectories and current local fiscal conditions. If either placebo produced significant second-stage results, it would indicate that the true Bartik IV is identifying a channel other than city-specific concentration dynamics—generic sector-shock exposure in the case of the permutation placebo, or slow-moving unobservables correlated with both long-run sector trends and local credit risk in the case of the future-shock placebo. In practice, neither placebo survives. The permutation placebo collapses the first-stage Kleibergen–Paap F to 3.7, and the future-shock placebo further collapses it to 1.3 and flips the sign of the HHI coefficient. Because both placebos fail to replicate the identification, the true Bartik IV loads on sector-specific concentration dynamics rather than on generic sector-shock exposure or persistent unobserved correlates.

V.G. Local Average Treatment Effect and Complier Characterization

The IV estimates identify a local average treatment effect (LATE) for cities whose concentration responds most to the Bartik instrument—that is, cities whose industrial composition is most sensitive to national sector-specific employment trends. The standard Rotemberg weight decomposition (Goldsmith-Pinkham, Sorkin, and Swift, 2020) formally characterizes which sectors drive a Bartik IV estimate, but applies only to linear shift-share instruments and is not available in closed form for the nonlinear predicted-HHI instrument used here. I instead characterize the LATE empirically: I rank cities by the within-city standard deviation of the Bartik predicted HHI and compare observable characteristics across terciles of

this distribution. Like the Rotemberg decomposition, this characterizes the variation that identifies the IV, but does so by sorting on cities rather than on sectors.

Complier cities—those in the top tercile of instrument variation—are smaller (median population 46,000 versus 63,000 for weak compliers), more concentrated (mean HHI 0.145 versus 0.106), and have lower property values (median \$172,000 versus \$202,000). They also exhibit more than twice as much within-city HHI volatility (standard deviation 0.015 versus 0.007), confirming that their industrial composition is more responsive to national sector trends. These are precisely the cities where concentration risk is most acute: they have fewer alternative employers, thinner fiscal buffers, and industrial mixes that are more exposed to sector-specific shocks.

The complier characterization helps explain why the IV point estimates exceed the OLS estimates. The LATE concentrates on cities where concentration changes have outsized fiscal consequences—smaller cities with lower property values and less diversified economies. For these cities, a given increase in HHI represents a more meaningful deterioration in fiscal risk than it would for a larger, more diversified city with substantial property-tax capacity. The OLS estimates, by contrast, reflect the average effect across all cities, including large diversified cities where HHI changes are small and fiscally inconsequential.

VI. ALTERNATIVE EXPLANATIONS

In this section, I consider several alternative explanations for the observed HHI–spread relationship.

VI.A. Systematic risk and city beta

I have argued that city bond spreads are primarily determined by idiosyncratic default risk. However, it is possible that the HHI coefficient could reflect unpriced variation in a city’s exposure to aggregate business-cycle risk rather than the idiosyncratic risk channel emphasized in the framework. In this section, I test this assumption by directly controlling for systematic risk exposure following Tuzel and Zhang (2017). For each NAICS supersector k , I estimate an industry GDP beta by regressing annual national sector employment growth on the national GDP growth rate:

$$(17) \quad \Delta \ln E_{kt}^{\text{nat}} = a_k + \lambda_k \Delta \ln \text{GDP}_t + \eta_{kt},$$

and construct a city-year beta as the employment-weighted average of the sector betas,

$$(18) \quad \hat{\beta}_{it} = \sum_{k=1}^K p_{kit} \hat{\lambda}_k.$$

The construction exactly mirrors the local-beta measure in Tuzel and Zhang (2017), the difference being that they aggregate to MSAs and I aggregate to cities. In the cross-section of cities, $\hat{\beta}_{it}$ and HHI are weakly correlated ($\rho \approx 0.08$), consistent with the observation that HHI measures the *dispersion* of a city’s employment portfolio while $\hat{\beta}$ measures the *identity* of its sectors (whether they are procyclical or countercyclical). The two are in principle orthogonal dimensions, and they turn out to be nearly so in the data.

I add $\hat{\beta}_{it}$ to the main specification in two ways. First, I include it as an exogenous control in both the OLS and IV regressions. Second, because $\hat{\beta}_{it}$ is itself a share-weighted function of the endogenous sector shares p_{kit} , I also instrument it with the Bartik-style shift-share version that replaces the contemporaneous shares with predetermined (2005) shares: $\hat{\beta}_{it}^{\text{pred}} = \sum_k p_{ki,0} \hat{\lambda}_k$. In the IV specification, the instrument set for the three endogenous variables (HHI, ln PV, $\hat{\beta}_{it}$) is the same set used in the main IV analysis.

Table IA.17 reports the results. In column (1), adding $\hat{\beta}_{it}$ as an exogenous control to the baseline OLS regression leaves the HHI coefficient essentially unchanged. In column (2), the same exercise in the IV specification yields similar stability, with the HHI coefficient moving by less than 5% of a standard deviation. These results indicate that this paper’s main estimates are not driven by unobserved heterogeneous exposure to aggregate risk.

VI.B. Local bias and liquidity

It is well-documented that municipal and non-profit bonds are disproportionately held by local retail investors, which leads to fragmented markets (Schultz, 2012; Dougal, Gao, Mayew, and Parsons, 2019; Babina, Jotikasthira, Lundblad, and Ramadorai, 2021). If investors in concentrated cities exhibit stronger local bias, perhaps because a dominant employer creates a more cohesive local identity, two opposing forces could be at work. On one hand, overoptimistic local investors would bid up prices and push yields down, which would attenuate the HHI–spread relationship and make my estimates a lower bound on the true credit-risk effect. On the other hand, concentrated local ownership could reduce secondary-market liquidity, inflating spreads through a liquidity premium rather than a default premium.

To distinguish these channels, Appendix C augments the baseline specification with four bond-level liquidity measures constructed from the trade data: a bid-ask proxy (Harris and Piwowar, 2006), yield dispersion (Jankowitsch, Nashikkar, and Subrahmanyam, 2011), the

zero-trading-day fraction (Chen, Lesmond, and Wei, 2007), and turnover. The HHI coefficient is stable—and if anything *increases*—when these controls are added, ruling out liquidity as the primary channel and suggesting that any local-bias effect works in the opposite direction, making the reported estimates conservative.

VI.C. Management quality

An alternative hypothesis is that concentrated cities might have weaker governance or less sophisticated fiscal management, and it is this mismanagement that raises borrowing costs, rather than industrial structure. While management quality is difficult to observe directly (Bloom and Van Reenen, 2007), several features of the research design address this concern. City fixed effects absorb persistent differences in governance quality across cities. Time-varying controls for population and unemployment capture changes in the scale of city operations. Most importantly, the Bartik instrument isolates variation in HHI driven by national industry employment trends, which are plausibly orthogonal to changes in local management quality.

VI.D. Institutional constraints

Most states require municipalities to maintain balanced budgets, and only about half of states authorize municipal Chapter 9 bankruptcy (Gao, Lee, and Murphy, 2019). These institutional features could affect the relationship between fiscal stress and borrowing costs. However, state bankruptcy authorization is time-invariant and is absorbed by city fixed effects. Internet Appendix G further shows that the results are robust to the inclusion of state $\times$ year fixed effects, which absorb all time-varying state-level policies including changes to fiscal oversight regimes.

VII. QUANTITATIVE MODEL

This section develops a quantitative model to rationalize the empirical results and to identify underlying structural parameters that govern the risk and agglomeration channels. The model is required for an identification reason that the reduced-form regressions cannot address: the observed spread combines a risk channel that raises spreads with an offsetting agglomeration cushion that lowers them, and these two channels cannot be separated by regression of spreads on HHI. By imposing structure on the loss distribution and the city’s fiscal cushion, the model backs out the gross magnitude of each channel: a 230 basis-point gross

risk effect of HHI, of which roughly two-thirds is offset by agglomeration in low-property-value cities and which is fully offset in high-property-value cities—a decomposition that is the central new result of this section and is not visible in the regression output alone.

The model is estimated in two stages. Stage 1 starts from the empirical observation that sector size is approximately log-normally distributed within a city, consistent with proportional random growth (Gibrat, 1931; Gabaix, 1999). Because pure Gibrat proportional growth generates a non-stationary distribution whose variance grows without bound, the model assumes dampened proportional sectoral growth in which log employment reverts to a long-run target (Kalecki, 1945). This framework is consistent with the well-documented empirical pattern that larger firms grow more slowly than smaller firms within an industry (Hall, 1987; Evans, 1987; Sutton, 1997).¹⁰ Stage 1 calibrates the dampened-growth process to match the cross-sectional HHI distribution and simultaneously generates an endogenous distribution of property values based on sector-level property value factors.

Stage 2 takes the simulated industrial structure as given and uses empirically observed city fiscal ratios to calibrate a fiscal default mechanism. The annual default probability is analytically converted into a credit spread following Duffie and Singleton (1999) and Lando (2004). The final result is the estimation of three structural parameters: latent fiscal capacity ψ , baseline default probability q_{base} , and an agglomeration factor γ .

VII.A. Stage 1: Industrial Dynamics and Endogenous Property Values

The industrial structure of the model is designed to target both long-run heterogeneity in sector size and cross-city heterogeneity in sector specialization. This allows the model to match the empirical observation that some sectors are persistently larger than others (e.g., Manufacturing vs. Mining), and that different cities have different long-run sectoral specializations (e.g., Manufacturing in Detroit and Tourism in Miami).

In particular, the model assumes that each city has K sectors, and the log employment of sector k in city c evolves as a dampened-growth AR(1) process,

$$(19) \quad \log E_{ck,t+1} = \log E_{ck,t} + \beta(m_{ck} - \log E_{ck,t}) + \sigma \varepsilon_{ck,t}, \quad \varepsilon_{ck,t} \sim \mathcal{N}(0, 1),$$

where β is the mean-reversion rate, σ is the per-sector annual shock standard deviation, and $m_{ck} = \mu_k + \tau \cdot \eta_{ck}$ is the city-specific long-run target for sector k . Here μ_k is the national long-run scale of sector k , $\eta_{ck} \sim \mathcal{N}(0, 1)$ is a city-specific deviation, and τ scales cross-city dispersion in long-run sector targets. The mean-reversion term $\beta(m - \log E_t)$ is

¹⁰An alternative method to generate stationarity is to allow entry and exit of sectors (Simon, 1955, 1960) which is less realistic in a study of just 20 coarse sectors.

the dampening that delivers a stationary cross-sectional distribution; setting $\beta = 0$ recovers the pure Gibrat law $\log E_{t+1} = \log E_t + \sigma\varepsilon$.

The same process determines a city's property values. Following a standard urban economics model (Rosen, 1979; Roback, 1982), the log of median property value is a share-weighted sum of sector-specific property-value premia v_k :

$$(20) \quad \ln(\text{PV}_c) = \alpha + \sum_{k=1}^K p_{ck} \cdot v_k.$$

Cities dominated by high-premium sectors (e.g., Information, Real Estate) have higher property values; cities dominated by low-premium sectors (e.g., Mining and Extraction, Utilities) have lower property values. In steady state, each city's employment shares $\mathbf{p}_c$ jointly determine its HHI and its property value.

VII.B. Stage 2: Fiscal Structure and Default

B.1. City Revenues and Expenses

Each city c collects revenue from two sources: property taxes (share ω_c of total revenue) and employment-sensitive revenue (share $1 - \omega_c$, which includes income and sales taxes, service fees, and intergovernmental transfers). The share ω_c is allowed to vary by city to reflect that cities with higher property values derive a larger share of revenue from property taxes (U.S. Census Bureau, 2017; Ahern, 2024). Pre-shock total revenue is normalized to unity.

Cities have two types of expenses: non-debt expenditure g_c (as a fraction of revenue) and debt service $d_c \cdot (r_f + s_c)$, where d_c is the leverage ratio (long-term debt to revenue), r_f is the risk-free rate, and s_c is the credit spread. Cities also maintain unrestricted reserve balances r_{base} as a buffer against revenue shocks. Non-debt expenditure and the leverage ratio are exogenous and constant across cities.

The assumption of exogenous fiscal ratios is analogous to structural corporate credit risk models (Merton, 1974; Leland, 1994) that take leverage as given when pricing credit risk, even though leverage is ultimately a choice variable. Similarly, though cities could adjust expenditure and leverage in response to employment shocks, the empirical evidence suggests that this margin is limited. Appendix Figure IA.1 plots expenditure shares, leverage ratios, and bond spreads by property-value tercile in the Census of Governments data and confirms that while spreads vary substantially across terciles, expenditure and leverage ratios are essentially flat. This is consistent with state-imposed balanced-budget requirements and debt limitations (Kiewiet and Szakaly, 1996; Epple and Spatt, 1986). Moreover, the composition of

fixed versus variable expenditures (personnel, infrastructure, debt service) is approximately constant across high- and low-property-value cities. Finally, Helm and Stuhler (2024) show that spending and taxing ratios are highly persistent, even after significant shocks.

B.2. Employment and property value shocks

Writing the one-period log-employment innovation as $\Delta_{ck} = -\beta(\log E_{ck} - m_{ck}) + \sigma\varepsilon_{ck}$, city-level employment growth is the share-weighted sum

$$(21) \quad \frac{\Delta E_c}{E_c} = \sum_{k=1}^K p_{ck} \cdot \Delta_{ck},$$

which implies that with iid shocks, $\text{Var}(\Delta E_c/E_c) = \sigma^2 \cdot \text{HHI}_c$.¹¹ The same sector shocks affect property values through equation (20): the post-shock log property value is $\ln(\text{PV}_c^{\text{post}}) = \alpha + \sum_k p_{ck}^{\text{post}} v_k$, where the post-shock shares p_{ck}^{post} are computed from the updated sector employment levels. A negative shock to a high-value sector (e.g., Information) causes a larger decline in $\text{PV}_c^{\text{post}}/\text{PV}_c^{\text{pre}}$ than a negative shock to a low-value sector, generating asymmetric property-value responses across cities of differing industrial composition.

B.3. Default probabilities and credit spreads

The annual revenue shortfall, as a fraction of initial annual revenue, is

$$(22) \quad \ell_c = -(1 - \omega_c) \cdot \frac{\Delta E_c}{E_c} - \omega_c \cdot \left(\frac{\text{PV}_c^{\text{post}}}{\text{PV}_c^{\text{pre}}} - 1 \right),$$

combining the loss of employment-sensitive revenue and the loss of property-tax revenue from erosion of the property tax base. Default occurs if the realized loss exceeds the city's fiscal cushion:

$$(23) \quad \ell_c > \underbrace{r_{\text{base}}}_{\text{baseline reserve}} + \underbrace{\psi \cdot \frac{\text{PV}_c}{\overline{\text{PV}}}}_{\text{latent fiscal capacity}} + \underbrace{\gamma \cdot \text{HHI}_c \cdot \frac{\text{PV}_c}{\overline{\text{PV}}}}_{\text{agglomeration capacity}}.$$

The latent fiscal capacity scales with the city's ex post property value relative to the cross-city mean, $\widetilde{\text{PV}}_c \equiv \text{PV}_c/\overline{\text{PV}}$, capturing the ability to raise additional revenue through assessment increases, special assessments, or rainy-day fund drawdowns. The agglomeration capacity is the additional fiscal-resilience benefit that accrues to cities through concentrated industrial

¹¹The nonlinear analog $\sum_k p_{ck}(\exp(\Delta_{ck}) - 1)$ introduces a Jensen drift independent of HHI_c . The magnitude of the gap is below 2 bps of revenue at the calibrated σ , well within the resolution of the targeted spread moments.

bases (Marshall, 1890; Duranton and Puga, 2004; Rosenthal and Strange, 2004). Agglomeration benefits accrue disproportionately in cities with high-skill, high-wage workers (Moretti, 2004, 2013; Diamond, 2016), and these productivity rents are capitalized into local property values (Roback, 1982; Glaeser, Gyourko, and Saks, 2006), so the same per-unit agglomeration capacity driven by concentration translates into larger fiscal resilience in wealthier cities. The baseline reserve is set uniformly across cities to focus identification on latent fiscal capacity and agglomeration.

The default condition generates an annual default probability $q_c^{\text{HHI}} = \Pr(\ell_c > \text{cushion}_c)$. A residual probability q_{base} captures non-modeled distress (pension underfunding, legal liabilities, political dysfunction), so that the total annual default probability under independent-events combination is $q_c = 1 - (1 - q_{\text{base}})(1 - q_c^{\text{HHI}})$.¹² Bondholders price municipal debt under the Duffie–Singleton recovery-of-market-value convention (Duffie and Singleton, 1999; Lando, 2004), which gives a closed-form spread $s_c = -\log(1 - q_c) \cdot (1 - \delta)$ that is independent of bond maturity.

B.4. The Model’s Mechanism

The model generates asymmetry between high- and low-property-value cities through three risk-side channels and one offsetting agglomeration channel. On the risk side, (i) *shock diversification* gives diversified cities lower revenue volatility—the net employment change has standard deviation proportional to $\sigma\sqrt{\text{HHI}_c}$, independent of property values; (ii) *property-tax stability* gives high-value cities a more stable revenue base because their property values are dominated by high-premium sectors and respond less to single-sector contractions (Lutz, Molloy, and Shan, 2011); and (iii) *latent fiscal capacity* gives high-value cities a larger emergency buffer through the $\psi \cdot \widetilde{\text{PV}}_c$ term, reflecting greater latent taxing capacity in wealthier jurisdictions.

The fourth channel, *agglomeration capacity*, partially offsets the first three. The term $\gamma \cdot \text{HHI}_c \cdot \widetilde{\text{PV}}_c$ gives concentrated cities a larger fiscal cushion through the productivity externalities documented in the urban economics literature. Critically, this channel is interactive with property value: the per-unit-HHI cushion gain is $\gamma \cdot \widetilde{\text{PV}}_c$, roughly three times larger in high-property-value cities ($\widetilde{\text{PV}} \approx 1.5$) than in low-property-value cities ($\widetilde{\text{PV}} \approx 0.5$). This asymmetric scaling implies that in low-property-value cities the risk channel is likely to dominate, while in high-property-value cities the agglomeration provides a stronger cushion to offset the increase in risk from higher concentration.

¹²The first-order approximation $q_c \approx q_{\text{base}} + q_c^{\text{HHI}}$ holds with error bounded by $q_{\text{base}} q_c^{\text{HHI}} < 0.001$ at the calibrated values.

VII.C. Parameters and Identification

C.1. Stage 1 Estimation

Stage 1 has two free parameters, σ and τ , calibrated to the cross-sectional mean and standard deviation of HHI by minimizing a weighted distance criterion across 5,000 simulated cities and a 200-period burn-in. The mean-reversion rate $\beta = 0.050$ is fixed externally from a panel AR(1) on log sector employment shares over 2002–2019, with city and sector fixed effects. The recovered values are $\sigma = 0.180$ and $\tau = 0.30$. Sector property-value loadings $\{v_k\}$ in equation (20) are fixed data inputs from a between-city OLS regression with $R^2 = 0.43$; the model-implied $\log(\text{PV})$ distribution is rescaled linearly so its mean and SD match the data exactly. The sector values range from a high of $v = 15.4$ (Information) to a low of $v = -3.6$ (Mining and Extraction).

Table V reports the fit of Stage 1. The model matches the body of the HHI distribution well (mean 0.130 vs. 0.120 in data, SD 0.038 vs. 0.039) but understates the extreme right tail (skewness 2.59 vs. 3.63), a familiar limitation of finite-variance specifications relative to Pareto-tailed concentration data; this leads to a conservative estimate of the gross risk channel.

Figure II plots the joint distribution of HHI and log property values. Both data and model share similar support and shape, centered near $\text{HHI} \approx 0.10$ and $\ln(\text{PV}) \approx 12.2$. The data show a modest negative correlation ($\rho = -0.094$); the city-specific long-run targets m_{ck} generate stronger sorting in the model ($\rho = -0.386$). The spread calibration conditions on realized property value through the tercile mean, so this deviation does not enter the targeted moments.

C.2. Stage 2 Estimation

Panel A of Table VI lists the fixed inputs in Stage 2: the property-tax share ω_c varies by tercile (0.292 in the high-property-value tercile vs. 0.177 in the low-property-value tercile), the risk-free rate is $r_f = 0.03$, and the recovery rate is $\delta = 0.40$ (Leland, 1994; Cornaggia, Cornaggia, and Hund, 2017). I draw 5,000 Monte Carlo realizations of one-year sector employment innovations and compute $q_c^{\text{HHI}} = \Pr(\ell_c > \text{cushion}_c)$ for each city. Crucially, the same σ governs both the cross-sectional HHI distribution in Stage 1 and the fiscal shocks driving default in Stage 2: the data on industrial structure directly discipline the default mechanism.

The three free parameters of Stage 2 are ψ , q_{base} , and γ , which are identified by three data moments from the OLS regressions: the low-property-value HHI gradient (45.1 bps),

TABLE V
Stage 1: Stochastic Sector Growth with Endogenous Property Values

	Model	Data
<i>Panel A: Industry dynamics</i>		
β (mean reversion, annual)	0.050	—
σ (sector shock SD, annual)	0.180	—
τ (city-magnet dispersion)	0.30	—
K (number of sectors)	20	20
<i>Panel B: Model fit</i>		
Mean of HHI	0.1301	0.1200
Standard deviation of HHI	0.0381	0.0393
Skewness of HHI	2.59	3.63
95th percentile of HHI	0.1994	0.1851
Mean of log(property value)	12.216	12.216
Standard deviation of log(property value)	0.596	0.596
Corr(HHI, log(property value))	−0.386	−0.094
<i>Panel C: Sector property-value loadings (from data, $R^2 = 0.43$)</i>		
Information	15.4	Fixed
Real Estate	10.6	Fixed
Professional and Business Services	5.7	Fixed
Agriculture	4.6	Fixed
Finance and Insurance	3.3	Fixed
...		
Arts, Entertainment, and Recreation	−0.3	Fixed
Utilities	−3.0	Fixed
Mining and Extraction	−3.6	Fixed
Intercept (α_{HV})	10.598	Fixed

Notes: Panel A reports the parameter values for the Stage 1 estimation. The mean-reversion rate β is fixed externally to its empirical value estimated from a panel AR(1) on log sector employment shares. The two free parameters, σ and τ , are calibrated jointly to match the cross-sectional mean and standard deviation of HHI. Panel B reports model fit statistics. Skewness and the 95th percentile of HHI are reported for diagnostic purposes but are not targeted in the calibration. Panel C reports property value capitalization factors.

the high-property-value HHI gradient (−34.8 bps), and the mean spread (19.5 bps). The model targets the OLS gradients (not the IV) because OLS represents the average partial effect for the full population while the IV regressions identify a complier-weighted LATE. A fourth parameter, $r_{\text{base}} = 0.05$, is fixed externally as discussed above; using $r_{\text{base}} = 0.10$ delivers a similar fit.

TABLE VI
Stage 2: Calibration of the Hazard-Rate Default Model

	Model	Data
<i>Panel A: Fiscal ratios (drawn from data, by property-value tercile)</i>		
<i>Low property value cities</i>		
Property tax share (ω_c)	0.177	Fixed
Expenditure ratio (g_c)	0.875	Fixed
Leverage ratio (d_c)	1.091	Fixed
<i>Middle property value cities</i>		
Property tax share (ω_c)	0.235	Fixed
Expenditure ratio (g_c)	0.870	Fixed
Leverage ratio (d_c)	1.185	Fixed
<i>High property value cities</i>		
Property tax share (ω_c)	0.292	Fixed
Expenditure ratio (g_c)	0.879	Fixed
Leverage ratio (d_c)	0.987	Fixed
Risk-free rate (r_f)	0.03	Fixed
Recovery rate (δ)	0.40	Fixed
<i>Panel B: In-sample model fit</i>		
Mean spread (bps)	19.1	19.5
HHI gradient, low PV cities (bps)	45.3	45.1
HHI gradient, high PV cities (bps)	-0.0	-34.8
<i>Panel C: Structural parameters</i>		
Baseline reserve (r_{base} , fixed)	0.050	—
Latent fiscal capacity (ψ)	0.080	—
Baseline default (q_{base})	0.0030	—
Agglomeration capacity (γ)	2.50	—
<i>Panel D: Channel decomposition (low PV cities, OLS slope)</i>		
Full-model HHI gradient (bps/HHI)	82	—
Risk channel only (cushion held constant)	230	—
Cushion side (loss distribution held constant)	-148	—
Cushion-side offset of the risk channel	64%	—

Notes: Panel A reports city-level fiscal ratios drawn from the Census of Governments fiscal panel conditional on the city's property-value tercile. Panel B presents the in-sample model fit. Panel C reports the estimated structural parameters. Panel D reports a city-pair Shapley decomposition that swaps each city's loss distribution and total cushion against those of paired cities within the low-property-value tercile, averaging the marginal contribution across pairs and bins.

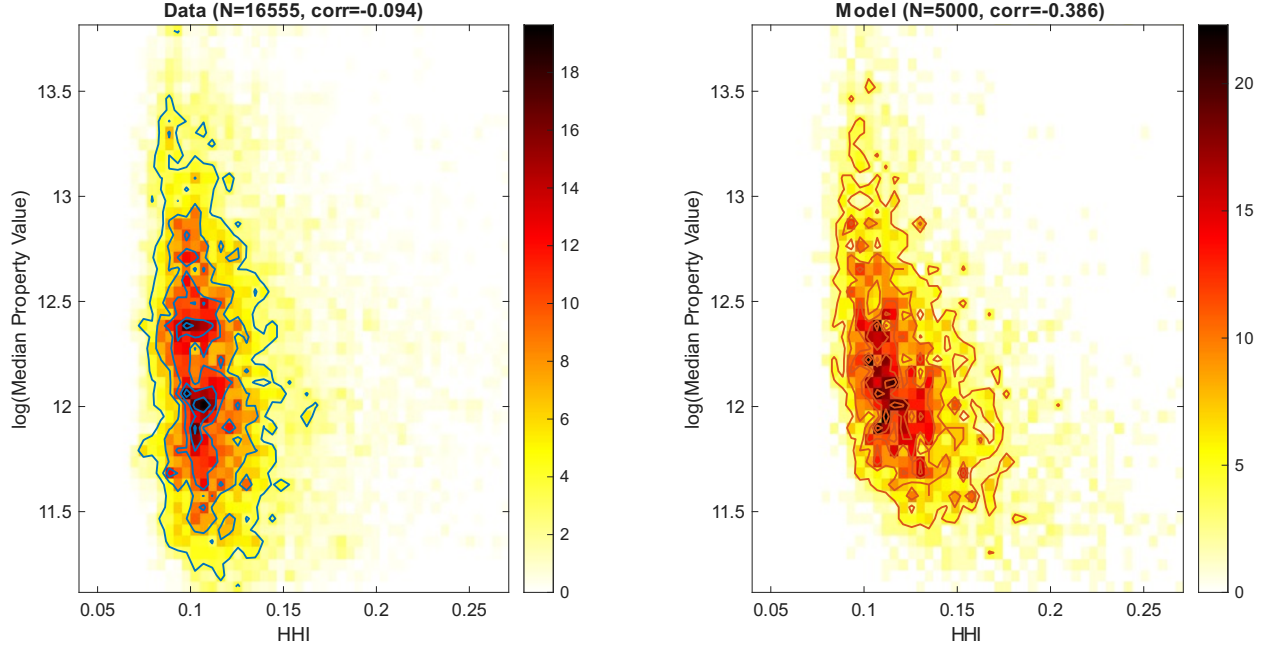


FIGURE II
Joint Distribution of HHI and Property Values: Data vs. Model

Notes: Bivariate heatmap of $\text{HHI} \times \log(\text{median property value})$ in the data (left, $N = 16,555$ city-years with non-missing HHI and ACS median home value used in the Stage 1 joint-distribution calibration; this is slightly smaller than the 18,447 city-year spread-regression sample because the latter has additional bond-trade observations not used in Stage 1) and the model (right, $N = 5,000$ cities, $\rho = -0.386$). Color intensity indicates density; contour lines show probability mass.

Each parameter is primarily identified by a different moment: q_{base} shifts the mean spread level; γ disciplines the gradient differential between high- and low-property-value cities through its interaction with $\widetilde{\text{PV}}$; ψ contributes a property-value-scaled level cushion to both terciles. Parameters are estimated by minimizing the sum of squared proportional deviations on a three-dimensional grid.

The model fits the Stage 2 moments closely on the level of mean spread and the low-property-value gradient, and produces a near-zero gradient in the high-property-value tercile. Table VI reports that the HHI gradient for low-property-value cities is 45.3 bps versus the 45.1 bps target, the mean spread is 19.1 bps versus 19.5, and the high-property-value gradient is -0.0 bps—consistent with the calibration target of approximately zero in the high-property-value tercile and within sampling noise of the (statistically null) point estimate of -34.8 bps in the data. Furthermore, Figure III plots the model-implied spread gradient against the data by HHI and property value tercile (left panel: low-property-value; right panel: high-property-value). The dotted line representing the full model closely follows the

data's positive relation between HHI and spreads in low-property-value cities and the neutral relationship for high-property-value cities.

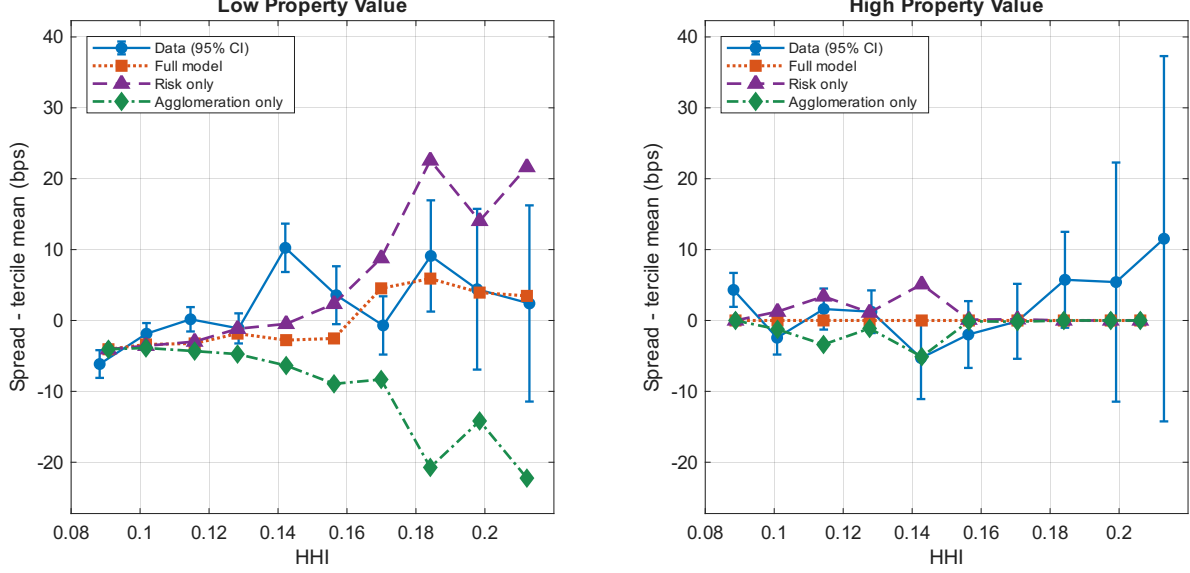


FIGURE III

Model Fit and Channel Decomposition: HHI–Spread Gradient by Property Value

Notes: Binned scatter of the HHI–spread relationship by property value tercile (low = left, high = right). Data points (solid line with circles) show the econometric prediction from the city and year fixed effects regression, demeaned within each property value tercile; error bars are 95% confidence intervals. The dotted line with squares shows the calibrated full model. The dashed line with triangles shows the risk-only counterfactual (city's own loss distribution paired with the within-tercile mean cushion); the dash-dotted line with diamonds shows the agglomeration-only counterfactual (city's own cushion paired with the within-tercile mean loss distribution). Both channel counterfactuals come from a city-pair Shapley decomposition.

VII.D. Model Results

Figure III plots the relationship between HHI and bond spreads separately for the risk and agglomeration channels. Within the low-property-value tercile, the full model generates an OLS HHI–spread gradient of 82 bps/HHI. A city-pair Shapley decomposition—which swaps each city's loss distribution and cushion against those of paired cities within the low-property-value tercile and averages the marginal contributions—attributes 230 bps/HHI to the risk side and -148 bps/HHI to the agglomeration channel, implying an agglomeration offset of roughly sixty-four percent of the gross risk (Panel D of Table VI and Figure III). I treat this decomposition as illustrative of the direction and rough magnitude of the agglomeration channel. The qualitative point is that the observed HHI–spread gradient understates the

gross risk from industrial concentration because agglomeration capacity partially insures concentrated cities.

Figure IV decomposes the annual fiscal loss and cushion by property-value tercile. The employment revenue loss is largest for low-property-value cities (which have a smaller property-tax buffer); the property tax erosion is largest for the high-property-value tercile. On the cushion side, the baseline reserve contributes a uniform $\sim 0.5\%$ per year, the latent fiscal capacity scales modestly with property value (from about 0.4% in the low-property-value tercile to about 1.8% in the high-property-value tercile), and the agglomeration capacity is the dominant differentiator, growing from approximately 1.7% in the low-property-value tercile to 6.8% in the high-property-value tercile.

At $\gamma^* = 2.50$, the agglomeration cushion contributes about 16% of annual revenue to the average low-property-value city versus about 64% to the average high-property-value city—a roughly four-to-one wedge driven entirely by the property-value multiplier and consistent in order of magnitude with the wage and property-value capitalizations documented in the spatial agglomeration literature (Moretti, 2013; Diamond, 2016). The latent fiscal capacity $\psi^* = 0.08$ contributes about 17% of revenue to the high-property-value cushion and 3% to the low-property-value cushion—a level effect that adds to, but does not dominate, the cross-tercile asymmetry. The baseline default probability $q_{\text{base}}^* = 0.0030$ contributes approximately 18 bps to the mean spread under the Duffie–Singleton hazard mapping.

D.1. Model Implications and Limitations

Within the limits of the stylized calibration, the model yields three implications that are not visible in the reduced-form and IV analyses on their own. First, the observed HHI–spread gradient understates the gross risk of industrial concentration: the risk channel alone would generate a steeper gradient than the observed one, with agglomeration capacity offsetting a meaningful share of that gross effect. Second, the reason the gradient is flat for high-property-value cities is not that concentration is harmless in these cities, but that agglomeration capacity and latent fiscal capacity proxied by property values nearly cancel the risk channel in wealthier cities—the net effect of concentration is close to zero because the opposing channels are similar in magnitude, not because either is small. Third, because agglomeration capacity scales multiplicatively with property value, poor concentrated cities bear the full gross risk while wealthy concentrated cities are insured by the same concentration that would otherwise expose them.

The model has several limitations. First, it is a partial-equilibrium model: industrial composition is taken as given from Stage 1, and the model cannot speak to whether cities *should*

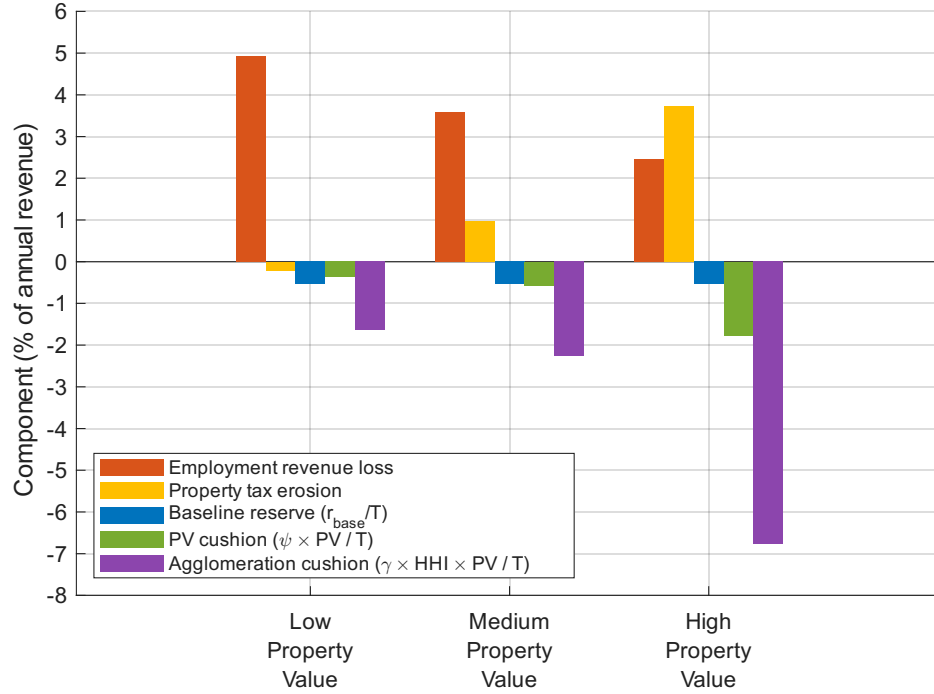


FIGURE IV
Annual Fiscal Loss Decomposition by Property Value Tercile

Notes: Decomposition of the annual fiscal loss and cushion into five components: employment revenue loss (red), property tax erosion (orange), baseline reserve r_{base} (blue, shown as negative = protection), latent fiscal capacity $\psi \cdot \widehat{\text{PV}}_c$ (green), and agglomeration capacity $\gamma \cdot \text{HHI}_c \cdot \widehat{\text{PV}}_c$ (purple). Values are percent of annual revenue. Loss components are conditional on a positive total revenue shortfall. Cushion components are annualized by dividing by $T = 9.4$ years (the par-weighted mean remaining maturity in the bond regression sample) for visual comparability with the per-year loss components; the underlying default mechanism remains one-period (annual).

specialize or diversify (the central question in general-equilibrium treatments such as Boehnert (2025)). Relatedly, γ is a reduced-form agglomeration-capacity parameter rather than a primitive in a structural production function. A fully structural treatment would require modeling agglomeration's wage and productivity effects directly. Second, the dampened-growth process slightly understates HHI skewness (model 2.59 vs. data 3.63), a familiar limitation of finite-variance specifications relative to Pareto-tailed concentration data; this leads to a conservative estimate of the gross risk channel.

VIII. CONCLUSION

This paper shows that industrial composition is priced into municipal borrowing costs. Using a panel of 1,177 U.S. cities from 2005 to 2022, I document that employment concentration raises credit spreads by approximately 4 basis points (per one-standard-deviation increase in HHI) among fiscally vulnerable, low-property-value cities, while having no observable effect on wealthy cities. To address potential endogeneity, I employ a two-channel instrumental variables strategy that utilizes a Bartik shift-share instrument for concentration and a housing-supply elasticity instrument for property values. These IV estimates confirm that the observed heterogeneity reflects a structural relationship: concentration causally increases spreads by amplifying idiosyncratic revenue risk, especially for cities with low property values.

To interpret these empirical patterns, I develop and calibrate a stylized model of stochastic industrial dynamics and fiscal default. The model decomposes the observed spread–HHI gradient into a gross concentration-risk channel and an offsetting agglomeration channel that property values capture. The calibrated gross risk effect is roughly three times the reduced-form estimate, with the agglomeration channel offsetting about two-thirds of it in low-property-value cities and the full effect in high-property-value cities. This decomposition explains why the reduced-form gradient is muted in wealthy cities even though the underlying risk channel is large.

The findings have implications for investors and policymakers. For investors, industrial concentration is a determinant of municipal credit spreads that is distinct from the shock-exposure channels documented in recent work and not captured by standard fiscal metrics. For policymakers, the results indicate that the marginal value of economic diversification for borrowing costs is largest in cities whose property-tax base is already thin. More broadly, the evidence identifies an unexamined margin in the cross-section of municipal credit risk that is shaped by the underlying structure of the local economy.

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